

Exploring Disparate Data: Part 2

Exploring Disparate Data: Part 3 - Final Report

Put your group name here: (anyname that is fancy and funny)

Due November 12th, 2025

Contents

0.1	Overview	2
0.2	NOAA data for Atlantic and Pacific Basin	3
0.3	Sea Ice Data	11
0.4	Stats and Plots	12
1	Opinions on climate data from Meta (from A2)	15
2	COVID Data from Our World In Data	16
3	World Happiness Report Score (from 2023 report)	29
4	Data Description	34
4.1	Climate Change Awareness Data	34
4.2	Covid 2020 Cases Data	34
4.3	World Happiness Report Data	35
4.4	Combining the Data	35
5	Exploratory Data Analysis	36
6	Conclusion and Future Work	38
6.1	Overview	39
6.2	Tubric	40

BEGIN ASSIGNMENT

```
knitr::opts_chunk$set(error = TRUE)
# DO NOT PUT install.packages() IN AN RMD!!!
# While you're here, make sure you group name is filled in above.
# (And make sure it ends with a quotation mark).
library(tidyverse)
```

```
## -- Attaching core tidyverse packages -----
## v dplyr      1.1.4      v readr      2.1.5
## v forcats    1.0.0      v stringr   1.5.2
## v ggplot2    4.0.0      v tibble    3.3.0
## v lubridate  1.9.4      v tidyr     1.3.1
## v purrr      1.1.0
## -- Conflicts -----
## x dplyr::filter() masks stats::filter()
## x dplyr::lag()    masks stats::lag()
## i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become errors
```

```
theme_set(theme_bw())
library(arrow)
```

```
##
## Attaching package: 'arrow'
##
## The following object is masked from 'package:lubridate':
##
##     duration
##
## The following object is masked from 'package:utils':
##
##     timestamp
```

```
library(openxlsx)
library(janitor)
```

```
##
## Attaching package: 'janitor'
##
## The following objects are masked from 'package:stats':
##
##     chisq.test, fisher.test
```

List all group members here (maximum 3 and minimum 2):

- Tran Tram Anh (Sophia) Thai
- Kaya Darling

This file will be submitted as an Rmd file to GradeScope. You will need to create a group on GradeScope with the correct members. *If the person who submits this does not list you, you will not get a grade on this assignment!!!* Make sure you trust the person who is submitting it!

0.1 Overview

This is **Part 1** of the group assignment, in which you'll demonstrate your ability to clean the data sets.

In this part, you will prepare the following data sets for analysis in Part 2. Note that you'll only need to use *three* of them, so you can focus on those *three* for cleaning and get started on Part 2.

Once you've completed the relevant sections, you will have the following files in your working directory:

1. `cyclones_data.parquet` contains data for hurricane strength in both the Atlantic and North Pacific basins.
2. `ice_extent_yearly.parquet` contains the yearly ice extent for the Arctic and the Antarctic poles.
3. `climate_awareness.parquet` contains the proportion of people from each country who answered “no”, “a little”, ... to a question asking about their awareness of the actual definitions of climate change.
4. `covid_2020.parquet` contains the total reported cases of COVID-19 in the year 2020 for all countries.
 - It also, quite helpfully, includes the continent on which each country sits, which can be joined with other data frames (with some care).
5. `happiness.parquet` contains information about the happiness of countries in the world, as measured by the `life_ladder` question (among other survey responses).

0.2 NOAA data for Atlantic and Pacific Basin

0.2.1 Load in the Data

In assignment 2, the NOAA data for the Atlantic was cleaned and tidied. That data will be useful for this assignment, so re-do the steps here. In this version, all of the steps can be done in a single pipeline. We'll also be doing the North Pacific (NP) basin.

```
cyclone_data_address <- "https://www.nhc.noaa.gov/data/hurdat/"
at_cyclone_filename <- "hurdat2-1851-2022-050423.txt"
np_cyclone_filename <- "hurdat2-nepac-1949-2022-050423.txt"

new_columns <- c("status", "latitude", "longitude", "max_wind",
  "min_pressure", "NE_extend_34", "SE_extend_34", "SW_extend_34",
  "NW_extend_34", "NE_extend_50", "SE_extend_50", "SW_extend_50",
  "NW_extend_50", "NE_extend_64", "SE_extend_64", "SW_extend_64",
  "NW_extend_64", "r_max_wind"
)

# Same steps as in A2, but you can put them all in the same pipeline!
at_cyclone <- str_c(cyclone_data_address, at_cyclone_filename, sep = "") |>
  read_csv(
    col_names = c(as.character(1:4)),
    progress = FALSE,
    show_col_types = FALSE
  ) |>
  separate_wider_delim(
    cols = `4`,
    # Set the delim and the names
    delim = ",",
    names = new_columns
  ) |>
  mutate(
    across(everything(), str_trim),
    # make "-999" NAs, make "-99" NAs
    # Create columns BasinNumberYear, Name, and Entries
    across(everything(), ~na_if(., "-999")),
    across(everything(), ~na_if(., "-99")),
    BasinNumberYear = ifelse(is.na(status), `1`, NA),
    Name = ifelse(is.na(status), `2`, NA),
    Entries = ifelse(is.na(status), `3`, NA)
```

```

) |>
relocate(BasinNumberYear, Name, Entries) |>
fill(BasinNumberYear, Name, Entries) |>
filter(!is.na(status)) |>
select(-Entries) |>
separate_wider_position(
  BasinNumberYear,
  # Specify the widths
  widths = c(Basin = 4, Number = 2, NameYear = 2)
) |>
separate_wider_position(
  `1`,
  # Specify the widths
  widths = c(ObservYear= 4, Month = 2, Day =2)
) |>
separate_wider_position(
  `2`,
  # Specify the widths
  widths = c(Hour = 2, Minute = 2)
) |>
rename(
  Identifier = `3`
) |>
mutate(
  across(
    c(NameYear, ObservYear, Month, Day, Hour,
      Minute, Number),
    as.integer
  )
) |>
mutate(across(max_wind:r_max_wind, as.numeric))

```

```

## Warning: One or more parsing issues, call 'problems()' on your data frame for details, e.g.:
##   dat <- vroom(...)
##   problems(dat)

```

at_cyclone

```

## # A tibble: 53,976 x 28
##   Basin Number NameYear Name   ObservYear Month   Day   Hour Minute Identifier status latitude longi
##   <chr>   <int>   <int> <chr>         <int> <int> <int> <int> <int> <chr>   <chr> <chr> <chr>
## 1 AL01     18     51 UNNAMED      1851     6    25     0     0 <NA>    HU    28.0N 94.8
## 2 AL01     18     51 UNNAMED      1851     6    25     6     0 <NA>    HU    28.0N 95.4
## 3 AL01     18     51 UNNAMED      1851     6    25    12     0 <NA>    HU    28.0N 96.0
## 4 AL01     18     51 UNNAMED      1851     6    25    18     0 <NA>    HU    28.1N 96.5
## 5 AL01     18     51 UNNAMED      1851     6    25    21     0 L      HU    28.2N 96.8
## 6 AL01     18     51 UNNAMED      1851     6    26     0     0 <NA>    HU    28.2N 97.0
## 7 AL01     18     51 UNNAMED      1851     6    26     6     0 <NA>    TS    28.3N 97.6
## 8 AL01     18     51 UNNAMED      1851     6    26    12     0 <NA>    TS    28.4N 98.3
## 9 AL01     18     51 UNNAMED      1851     6    26    18     0 <NA>    TS    28.6N 98.9
## 10 AL01    18     51 UNNAMED      1851     6    27     0     0 <NA>    TS    29.0N 99.4
## # i 53,966 more rows
## # i 12 more variables: SE_extend_34 <dbl>, SW_extend_34 <dbl>, NW_extend_34 <dbl>, NE_extend_50 <dbl>

```

```
## # SW_extend_50 <dbl>, NW_extend_50 <dbl>, NE_extend_64 <dbl>, SE_extend_64 <dbl>, SW_extend_64 <dbl>
```

```
# print(at_cyclone)
```

```
. = ottr::check("tests/at_cyclone.R")
```

```
## Warning in file(filename, "r"): cannot open file 'tests/at_cyclone.R': No such file or directory
```

```
## Error in file(filename, "r"): cannot open the connection
```

```
np_cyclone <- str_c(cyclone_data_address, np_cyclone_filename, sep = "") |>
  # ALL of the steps all over again
  # YOUR CODE HERE
  read_csv(
    col_names = c(as.character(1:4)),
    progress = FALSE,
    show_col_types = FALSE
  ) |>
  separate_wider_delim(
    cols = `4`,
    # Set the delim and the names
    delim = ",",
    names = new_columns
  ) |>
  mutate(
    across(everything(), str_trim),
    # make "-999" NAs, make "-99" NAs
    # Create columns BasinNumberYear, Name, and Entries
    across(everything(), ~na_if(., "-999")),
    across(everything(), ~na_if(., "-99")),
    BasinNumberYear = ifelse(is.na(status), `1`, NA),
    Name = ifelse(is.na(status), `2`, NA),
    Entries = ifelse(is.na(status), `3`, NA)
  ) |>
  relocate(BasinNumberYear, Name, Entries) |>
  fill(BasinNumberYear, Name, Entries) |>
  filter(!is.na(status)) |>
  select(-Entries) |>
  separate_wider_position(
    BasinNumberYear,
    # Specify the widths
    widths = c(Basin = 4, Number = 2, NameYear = 2)
  ) |>
  separate_wider_position(
    `1`,
    # Specify the widths
    widths = c(ObservYear= 4, Month = 2, Day =2)
  ) |>
  separate_wider_position(
    `2`,
    # Specify the widths
    widths = c(Hour = 2, Minute = 2)
```

```

) |>
  rename(
    Identifier = `3`
  ) |>
  mutate(
    across(
      c(NameYear, ObservYear, Month, Day, Hour,
        Minute, Number),
      as.integer
    )
  ) |>
  mutate(across(max_wind:r_max_wind, as.numeric))

```

```

## Warning: One or more parsing issues, call 'problems()' on your data frame for details, e.g.:
##   dat <- vroom(...)
##   problems(dat)

```

```
np_cyclone
```

```

## # A tibble: 30,625 x 28
##   Basin Number NameYear Name   ObservYear Month   Day   Hour Minute Identifier status latitude longitude
##   <chr>   <int>   <int> <chr>         <int> <int> <int> <int> <int> <chr>   <chr> <chr> <chr>
## 1 EP01     19     49 UNNAMED    1949     6    11     0     0 <NA>    TS    20.2N 106.3E
## 2 EP01     19     49 UNNAMED    1949     6    11     6     0 <NA>    TS    20.2N 106.4E
## 3 EP01     19     49 UNNAMED    1949     6    11    12     0 <NA>    TS    20.2N 106.5E
## 4 EP01     19     49 UNNAMED    1949     6    11    18     0 <NA>    TS    20.3N 107.0E
## 5 EP01     19     49 UNNAMED    1949     6    12     0     0 <NA>    TS    20.4N 108.0E
## 6 EP01     19     49 UNNAMED    1949     6    12     6     0 <NA>    TS    20.5N 109.0E
## 7 EP01     19     49 UNNAMED    1949     6    12    12     0 <NA>    TS    20.6N 110.0E
## 8 EP02     19     49 UNNAMED    1949     6    17    12     0 <NA>    TS    14.1N 107.0E
## 9 EP02     19     49 UNNAMED    1949     6    17    18     0 <NA>    TS    14.6N 107.5E
## 10 EP02    19     49 UNNAMED    1949     6    18     0     0 <NA>    TS    15.0N 108.0E
## # i 30,615 more rows
## # i 12 more variables: SE_extend_34 <dbl>, SW_extend_34 <dbl>, NW_extend_34 <dbl>, NE_extend_50 <dbl>,
## #   SW_extend_50 <dbl>, NW_extend_50 <dbl>, NE_extend_64 <dbl>, SE_extend_64 <dbl>, SW_extend_64 <dbl>

```

```
print(np_cyclone)
```

```

## # A tibble: 30,625 x 28
##   Basin Number NameYear Name   ObservYear Month   Day   Hour Minute Identifier status latitude longitude
##   <chr>   <int>   <int> <chr>         <int> <int> <int> <int> <int> <chr>   <chr> <chr> <chr>
## 1 EP01     19     49 UNNAMED    1949     6    11     0     0 <NA>    TS    20.2N 106.3E
## 2 EP01     19     49 UNNAMED    1949     6    11     6     0 <NA>    TS    20.2N 106.4E
## 3 EP01     19     49 UNNAMED    1949     6    11    12     0 <NA>    TS    20.2N 106.5E
## 4 EP01     19     49 UNNAMED    1949     6    11    18     0 <NA>    TS    20.3N 107.0E
## 5 EP01     19     49 UNNAMED    1949     6    12     0     0 <NA>    TS    20.4N 108.0E
## 6 EP01     19     49 UNNAMED    1949     6    12     6     0 <NA>    TS    20.5N 109.0E
## 7 EP01     19     49 UNNAMED    1949     6    12    12     0 <NA>    TS    20.6N 110.0E
## 8 EP02     19     49 UNNAMED    1949     6    17    12     0 <NA>    TS    14.1N 107.0E
## 9 EP02     19     49 UNNAMED    1949     6    17    18     0 <NA>    TS    14.6N 107.5E
## 10 EP02    19     49 UNNAMED    1949     6    18     0     0 <NA>    TS    15.0N 108.0E
## # i 30,615 more rows

```

```
## # i 12 more variables: SE_extend_34 <dbl>, SW_extend_34 <dbl>, NW_extend_34 <dbl>, NE_extend_50 <dbl>
## #   SW_extend_50 <dbl>, NW_extend_50 <dbl>, NE_extend_64 <dbl>, SE_extend_64 <dbl>, SW_extend_64 <dbl>
```

```
. = ottr::check("tests/np_cyclone.R")
```

```
## Warning in file(filename, "r"): cannot open file 'tests/np_cyclone.R': No such file or directory
```

```
## Error in file(filename, "r"): cannot open the connection
```

0.2.2 Combine into One Data Frame

Now, bind the two data frames together (row-wise). Since the **Basin** column already contains the information about which basin the cyclone was in, there's no need to specify an ID. (Hint: don't over think this one.)

Name the data frame `cyclones_data_update_0`.

```
# YOUR CODE HERE
cyclones_data_update_0 <- bind_rows(at_cyclone, np_cyclone)

print(cyclones_data_update_0)
```

```
## # A tibble: 84,601 x 28
##   Basin Number NameYear Name      ObservYear Month   Day   Hour Minute Identifier status latitude longitude
##   <chr>   <int>   <int> <chr>         <int> <int> <int> <int> <int> <chr>      <chr>   <chr>   <chr>
## 1 AL01      18      51 UNNAMED      1851     6    25     0     0 <NA>      HU      28.0N   94.8W
## 2 AL01      18      51 UNNAMED      1851     6    25     6     0 <NA>      HU      28.0N   95.4W
## 3 AL01      18      51 UNNAMED      1851     6    25    12     0 <NA>      HU      28.0N   96.0W
## 4 AL01      18      51 UNNAMED      1851     6    25    18     0 <NA>      HU      28.1N   96.5W
## 5 AL01      18      51 UNNAMED      1851     6    25    21     0 L        HU      28.2N   96.8W
## 6 AL01      18      51 UNNAMED      1851     6    26     0     0 <NA>      HU      28.2N   97.0W
## 7 AL01      18      51 UNNAMED      1851     6    26     6     0 <NA>      TS      28.3N   97.6W
## 8 AL01      18      51 UNNAMED      1851     6    26    12     0 <NA>      TS      28.4N   98.3W
## 9 AL01      18      51 UNNAMED      1851     6    26    18     0 <NA>      TS      28.6N   98.9W
## 10 AL01     18      51 UNNAMED      1851     6    27     0     0 <NA>      TS      29.0N   99.4W
## # i 84,591 more rows
## # i 12 more variables: SE_extend_34 <dbl>, SW_extend_34 <dbl>, NW_extend_34 <dbl>, NE_extend_50 <dbl>
## #   SW_extend_50 <dbl>, NW_extend_50 <dbl>, NE_extend_64 <dbl>, SE_extend_64 <dbl>, SW_extend_64 <dbl>
```

```
. = ottr::check("tests/cyclones_data_update_1.R")
```

```
## Warning in file(filename, "r"): cannot open file 'tests/cyclones_data_update_1.R': No such file or d
```

```
## Error in file(filename, "r"): cannot open the connection
```

0.2.3 Fix Latitude and Longitude

The latitude and longitude are not in a format that makes for nice plotting. They're encoded as "28.0N" for 28 degrees North and "28.0S" for 28 degrees South of the equator. By convention, "28.0N" should be 28 degrees (a positive number) and 28.8S should be -28 degrees (a negative number). Similarly, 94W should be -94 while 94E should be 94. Fix this.

Write a function that takes in a vector of latitude *or* longitude in the format of these data, then outputs a positive version of the numeric part if there's an "N" or an "E", and a negative version of the numeric part if there's a "W" or "S".

There are many ways to solve this. My solution is one line and uses `parse_number()` and a single `if_else()` statement with a regex, but there are many possible ways to do this!

```
convert_latlon <- function(latlon) {
  # YOUR CODE HERE
  if_else(
    str_detect(latlon, "[NS]$"), # check if N/S for latitude
    parse_number(latlon) * if_else(str_detect(latlon, "N$"), 1, -1),
    parse_number(latlon) * if_else(str_detect(latlon, "E$"), 1, -1)
  )
}

test_data <- c("49W", "49.99W", "49E", "49.99E", "49N", "49.99S", "-0.0W")

# If this returns "TRUE", then your function will work on the real data.
all.equal(
  convert_latlon(test_data),
  c(-49, -49.99, 49, 49.99, 49, -49.99, 0)
)
```

```
## [1] TRUE
```

```
cyclones_data_update_1 <- cyclones_data_update_0 |>
  mutate(
    lat = convert_latlon(latitude),
    lon = convert_latlon(longitude)
  )
```

```
. = ottr::check("tests/convert_latlon.R")
```

```
## Warning in file(filename, "r"): cannot open file 'tests/convert_latlon.R': No such file or directory
```

```
## Error in file(filename, "r"): cannot open the connection
```

0.2.4 Extract Dates

We have a few more tasks. First, create a datetime object using the `make_datetime()` function so that R knows how to plot the times. Label the new column as "date".

To do this, you must choose whether to use `ObservYear` or `NameYear`.

```
cyclones_data_update_2 <- cyclones_data_update_1 |>
  mutate(
    # YOUR CODE HERE
    NameYear = as.integer(NameYear),
    Month = as.integer(Month),
    Day = as.integer(Day),
```



```

    Hour = as.integer(Hour),
    date = make_datetime(
      year = NameYear,
      month = Month,
      day = Day,
      hour = Hour
    )
  )
)

print(cyclones_data_update_2)

```

```

## # A tibble: 84,601 x 31
##   Basin Number NameYear Name      ObservYear Month   Day   Hour Minute Identifier status latitude longitude
##   <chr>   <int>   <int> <chr>         <int> <int> <int> <int> <int> <chr>      <chr> <chr>   <chr>
## 1 AL01      18      51 UNNAMED      1851     6    25     0     0 <NA>      HU      28.0N  94.8W
## 2 AL01      18      51 UNNAMED      1851     6    25     6     0 <NA>      HU      28.0N  95.4W
## 3 AL01      18      51 UNNAMED      1851     6    25    12     0 <NA>      HU      28.0N  96.0W
## 4 AL01      18      51 UNNAMED      1851     6    25    18     0 <NA>      HU      28.1N  96.5W
## 5 AL01      18      51 UNNAMED      1851     6    25    21     0 L          HU      28.2N  96.8W
## 6 AL01      18      51 UNNAMED      1851     6    26     0     0 <NA>      HU      28.2N  97.0W
## 7 AL01      18      51 UNNAMED      1851     6    26     6     0 <NA>      TS      28.3N  97.6W
## 8 AL01      18      51 UNNAMED      1851     6    26    12     0 <NA>      TS      28.4N  98.3W
## 9 AL01      18      51 UNNAMED      1851     6    26    18     0 <NA>      TS      28.6N  98.9W
## 10 AL01     18      51 UNNAMED      1851     6    27     0     0 <NA>      TS      29.0N  99.4W
## # i 84,591 more rows
## # i 15 more variables: SE_extend_34 <dbl>, SW_extend_34 <dbl>, NW_extend_34 <dbl>, NE_extend_50 <dbl>,
## #   SW_extend_50 <dbl>, NW_extend_50 <dbl>, NE_extend_64 <dbl>, SE_extend_64 <dbl>, SW_extend_64 <dbl>,
## #   r_max_wind <dbl>, lat <dbl>, lon <dbl>, date <dtm>

```

```

. = ottr::check("tests/cyclones_data_update_2.R")

```

```

## Warning in file(filename, "r"): cannot open file 'tests/cyclones_data_update_2.R': No such file or directory
## Error in file(filename, "r"): cannot open the connection

```

0.2.5 Assign Storm Categories

Unlike in assignment 1, the category is not part of these data. However, we learned how storms are categorized! In particular, meteorologists use the Saffir–Simpson hurricane wind scale (SSHWS) (note that our data are measured in knots on that scale).

Categorize the cyclones based on the scale above using a `case_when()` statement based on the `max_wind` column. The code is set up to create an *ordered factor* so that R knows that the levels are ordered as TD < TS < 1 < 2 < 3 < 4 < 5.

```

cat_levels <- c("TD", "TS", "1", "2", "3", "4", "5")

cyclones_data <- cyclones_data_update_2 |>
  mutate(
    category = ordered(
      case_when(
        # YOUR CODE HERE

```

```

        max_wind < 34 ~ "TD",
        max_wind < 64 ~ "TS",
        max_wind < 83 ~ "1",
        max_wind < 96 ~ "2",
        max_wind < 113 ~ "3",
        max_wind < 137 ~ "4",
        max_wind >= 137 ~ "5"
    ),
    levels = cat_levels
)
)

print(cyclones_data)

```

```

## # A tibble: 84,601 x 32
##   Basin Number NameYear Name   ObservYear Month   Day   Hour Minute Identifier status latitude longitude
##   <chr>   <int>   <int> <chr>         <int> <int> <int> <int> <int> <chr>   <chr> <chr>   <chr>
## 1 AL01     18     51 UNNAMED      1851     6    25     0     0 <NA>    HU    28.0N  94.8W
## 2 AL01     18     51 UNNAMED      1851     6    25     6     0 <NA>    HU    28.0N  95.4W
## 3 AL01     18     51 UNNAMED      1851     6    25    12     0 <NA>    HU    28.0N  96.0W
## 4 AL01     18     51 UNNAMED      1851     6    25    18     0 <NA>    HU    28.1N  96.5W
## 5 AL01     18     51 UNNAMED      1851     6    25    21     0 L      HU    28.2N  96.8W
## 6 AL01     18     51 UNNAMED      1851     6    26     0     0 <NA>    HU    28.2N  97.0W
## 7 AL01     18     51 UNNAMED      1851     6    26     6     0 <NA>    TS    28.3N  97.6W
## 8 AL01     18     51 UNNAMED      1851     6    26    12     0 <NA>    TS    28.4N  98.3W
## 9 AL01     18     51 UNNAMED      1851     6    26    18     0 <NA>    TS    28.6N  98.9W
## 10 AL01    18     51 UNNAMED      1851     6    27     0     0 <NA>    TS    29.0N  99.4W
## # i 84,591 more rows
## # i 16 more variables: SE_extend_34 <dbl>, SW_extend_34 <dbl>, NW_extend_34 <dbl>, NE_extend_50 <dbl>,
## #   SW_extend_50 <dbl>, NW_extend_50 <dbl>, NE_extend_64 <dbl>, SE_extend_64 <dbl>, SW_extend_64 <dbl>,
## #   r_max_wind <dbl>, lat <dbl>, lon <dbl>, date <dtm>, category <ord>

```

```

. = ottr::check("tests/cyclones_update_2.R")

```

```

## Warning in file(filename, "r"): cannot open file 'tests/cyclones_update_2.R': No such file or directory

```

```

## Error in file(filename, "r"): cannot open the connection

```

0.2.6 Plots and Summary Statistics!

0.2.7 Clean Up

Now that we have a suitable data frame, save it as a parquet file with the file name "cyclones_data.parquet".

```

# YOUR CODE HERE
write_parquet(cyclones_data, "cyclones_data.parquet")

```

```

. = ottr::check("tests/write_cyclones.R")

```

```

## Warning in file(filename, "r"): cannot open file 'tests/write_cyclones.R': No such file or directory

```

```

## Error in file(filename, "r"): cannot open the connection

```

0.3 Sea Ice Data

The following data are used in the Course Notes to demonstrate reading in specific sheets. In this assignment, we care most about what's *in* the data! I've copied the code over from the notes; there are no questions on loading the data.

The sea ice extent is defined as the total area of the ocean that is covered in ice of at least 15% concentration.

```
sea_ice_extent_xlsx <- "https://masie_web.apps.nsidc.org/pub//DATASETS/NOAA/G02135/seaice_analysis/Sea_
```

```
NH_daily <- sea_ice_extent_xlsx |>
  read.xlsx(
    sheet = "NH-Daily-Extent",
  ) |>
  select(X1, X2, `1978`:`2023`) |>
  rename(
    month = X1,
    day = X2
  ) |>
  fill(month) |>
  pivot_longer(
    cols = `1978`:`2023`,
    names_to = "year",
    values_to = "ice_extent",
    values_drop_na = TRUE,
  ) |>
  mutate(
    year = as.integer(year),
    month = ordered(
      month,
      levels = c("January", "February", "March", "April",
                  "May", "June", "July", "August", "September",
                  "October", "November", "December")),
    region = "Arctic",
  ) |>
  arrange(
    year, month, day
  )
```

```
SH_daily <- sea_ice_extent_xlsx |>
  read.xlsx(
    sheet = "SH-Daily-Extent",
    skipEmptyCols = TRUE,
    fillMergedCells = TRUE,
    cols = 1:48
  ) |>
  rename(
    month = X1,
    day = X2
  ) |>
  pivot_longer(
    cols = `1978`:`2023`,
    names_to = "year",
    names_transform = list(year = as.integer),
```

```

    values_to = "ice_extent",
    values_drop_na = TRUE,
  ) |>
  mutate(
    month = ordered(
      month,
      levels = c("January", "February", "March", "April",
                  "May", "June", "July", "August", "September",
                  "October", "November", "December")
    ),
    region = "Antarctic",
  ) |>
  arrange(
    year, month, day
  )

```

```

ice_extent_daily <- bind_rows(NH_daily, SH_daily) |>
  mutate(date = make_date(year, month, day)) |>
  arrange(region, date)

```

```
ice_extent_daily
```

```

## # A tibble: 29,704 x 6
##   month      day year ice_extent region    date
##   <ord>    <dbl> <int>      <dbl> <chr>    <date>
## 1 October      26  1978      17.6 Antarctic 1978-10-26
## 2 October      28  1978      17.8 Antarctic 1978-10-28
## 3 October      30  1978      17.7 Antarctic 1978-10-30
## 4 November       1  1978      17.5 Antarctic 1978-11-01
## 5 November       3  1978      17.5 Antarctic 1978-11-03
## 6 November       5  1978      17.3 Antarctic 1978-11-05
## 7 November       7  1978      17.2 Antarctic 1978-11-07
## 8 November       9  1978      17.0 Antarctic 1978-11-09
## 9 November      11  1978      16.8 Antarctic 1978-11-11
## 10 November     13  1978      16.4 Antarctic 1978-11-13
## # i 29,694 more rows

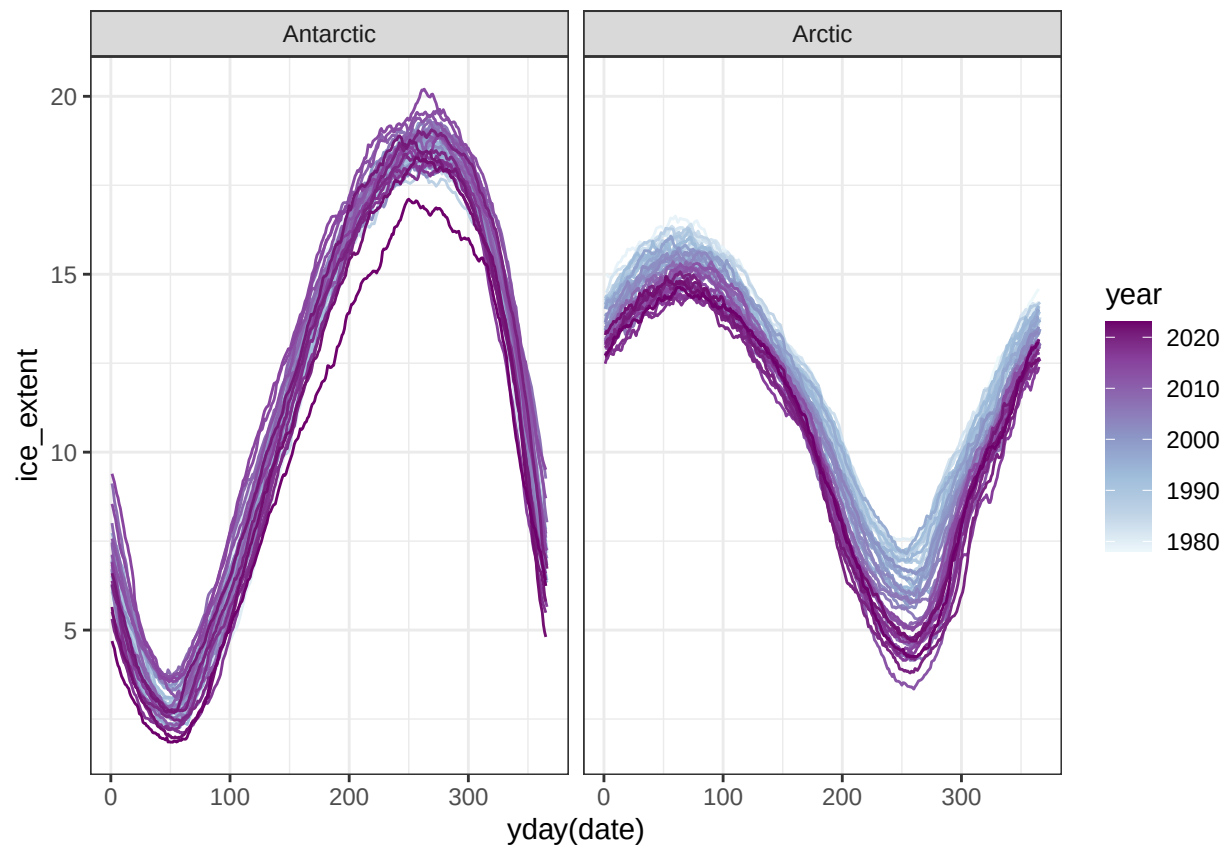
```

0.4 Stats and Plots

```

ice_extent_daily |>
  ggplot() +
    aes(x = yday(date), y = ice_extent, colour = year, group = factor(year)) +
    geom_line() +
    facet_wrap(~region) +
    #coord_polar() +
    scale_colour_distiller(
      direction = 1, type = "seq", palette = 3
    )

```



In the chunk below, find the maximum and minimum sea ice extent for each year (by region). Pivot the data frame so that there's a column labelled "name" and a column labelled "value", as below.

year	region	name	value
1978	Antarctic	min	7.283
1978	Antarctic	max	17.803
1978	Arctic	min	10.231
1978	Arctic	max	14.585
1979	Antarctic	min	2.911
1979	Antarctic	max	18.361
...	...	...	...

```
ice_extent_yearly <- ice_extent_daily |>
  # YOUR CODE HERE
  group_by(year, region) |>
  summarise(
    min = min(ice_extent, na.rm = TRUE),
    max = max(ice_extent, na.rm = TRUE),
    .groups = "drop"
  ) |>
  pivot_longer(
    cols = c(min, max),
    names_to = "name",
    values_to = "value"
  ) |>
```

```
mutate(name = as.character(name)) |>
  arrange(year, region, factor(name, levels = c("min", "max")))

ice_extent_yearly
```

```
## # A tibble: 184 x 4
##   year region    name value
##   <int> <chr>    <chr> <dbl>
## 1  1978 Antarctic min    7.28
## 2  1978 Antarctic max   17.8
## 3  1978 Arctic    min   10.2
## 4  1978 Arctic    max   14.6
## 5  1979 Antarctic min    2.91
## 6  1979 Antarctic max   18.4
## 7  1979 Arctic    min    6.90
## 8  1979 Arctic    max   16.6
## 9  1980 Antarctic min    2.52
## 10 1980 Antarctic max   19.1
## # i 174 more rows
```

```
. = ottr::check("tests/ice_extent_yearly.R")
```

```
## Warning in file(filename, "r"): cannot open file 'tests/ice_extent_yearly.R': No such file or directory
```

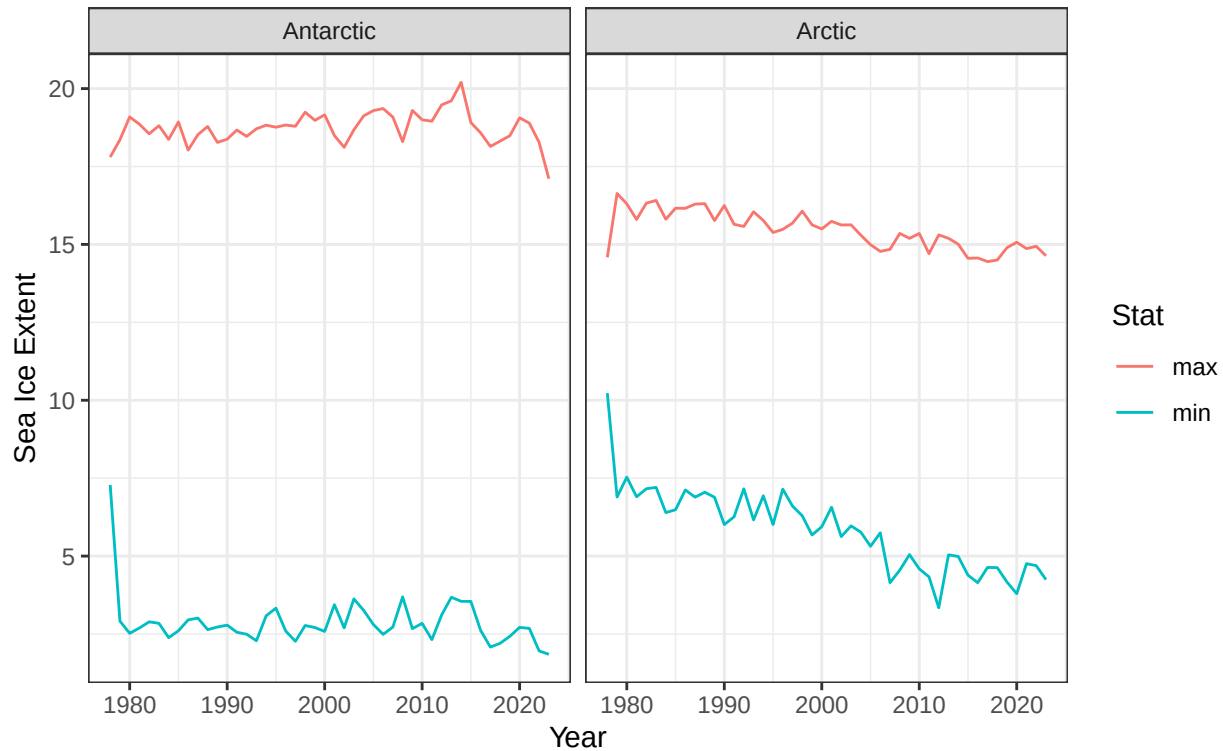
```
## Error in file(filename, "r"): cannot open the connection
```

Given your answer, the following code should produce a plot that's easier to interpret than the one before.

```
ggplot(ice_extent_yearly) +
  aes(x = year, y = value, colour = name) +
  geom_line() +
  facet_wrap(~ region) +
  labs(
    x = "Year", y = "Sea Ice Extent",
    colour = "Stat",
    title = "Min and Max Sea Ice Extent, by Year",
    subtitle = "Arctic is clearly decreasing, Antarctic is possibly becoming more variable."
  )
```

Min and Max Sea Ice Extent, by Year

Arctic is clearly decreasing, Antarctic is possibly becoming more variable.



Finally, let's save the data to `ice_extent_yearly.parquet`.

```
# YOUR CODE HERE
write_parquet(ice_extent_yearly, "ice_extent_yearly.parquet")
```

```
. = ottr::check("tests/sea_ice_parquet.R")
```

```
## Warning in file(filename, "r"): cannot open file 'tests/sea_ice_parquet.R': No such file or directory
```

```
## Error in file(filename, "r"): cannot open the connection
```

1 Opinions on climate data from Meta (from A2)

These data are the same as the data we worked with on A2. The following code will load in the awareness data; you are welcome to copy it for the happening data for the purpose of this assignment (you will not be graded on this, but it may benefit you in part 2).

```
climate_opinion_address <- "https://data.humdata.org/dataset/dc9f2ca4-8b62-4747-89b1-db426ce617a0/resou

climate_sheet_names <- climate_opinion_address |>
  loadWorkbook() |>
  names()
```

```

aware_sheet_name <- "climate_awareness"

climate_awareness <- climate_opinion_address |>
  read.xlsx(
    sheet = aware_sheet_name
  ) |>
  pivot_longer(
    cols = !contains(aware_sheet_name),
    names_to = "country",
    values_to = "score"
  ) |>
  mutate(
    climate_awareness = case_when(
      climate_awareness == "I have never heard of it" ~ "aware_no",
      climate_awareness == "I know a little about it" ~ "aware_alittle",
      climate_awareness == "I know a moderate amount about it" ~
        "aware_moderate",
      climate_awareness == "I know a lot about it" ~ "aware_alot",
      climate_awareness == "Refused" ~ "aware_refuse",
      climate_awareness == "(Unweighted Base)" ~ "aware_base"
    )
  ) |>
  rename(answer = climate_awareness) |>
  pivot_wider(
    names_from = answer,
    values_from = score
  )

write_parquet(climate_awareness, "climate_awareness.parquet")

climate_awareness

```

```

## # A tibble: 110 x 7
##   country          aware_no aware_alittle aware_moderate aware_alot aware_refuse aware_base
##   <chr>          <dbl>      <dbl>          <dbl>      <dbl>      <dbl>      <dbl>
## 1 Albania         9.99        43.0           33.1       10.6        3.25       329
## 2 Algeria        23.2         37.6           26.0       11.1        2.15      1066
## 3 Angola         19.1         43.0           23.5       11.1        3.29       727
## 4 Argentina       8.25         37.4           43.8        9.88       0.696     1257
## 5 Armenia         9.18         51.4           25.8       11.4        2.14       306
## 6 Asian.&.Pacific.Islands 13.3         40.1           24.8       19.5        2.26     1916
## 7 Australia        1.13         26.4           48.1       23.9        0.407     1012
## 8 Austria          1.34         16.4           48.6       32.7        0.985     1120
## 9 Azerbaijan       6.19         42.4           35.9       13.2        2.20       541
## 10 Bangladesh     24.5         36.2           16.2       18.9        4.22       910
## # i 100 more rows

```

2 COVID Data from Our World In Data

Perhaps there's some relationship between governments' response to the pandemic and governments' response to climate change? The following data will load in some information about COVID-19 in countries around the world.


```
#owid_address <- "https://covid.ourworldindata.org/data/owid-covid-data.csv"

owid_address <- "https://raw.githubusercontent.com/owid/covid-19-data/refs/heads/master/public/data/owid-covid-data.csv"

covid_cases <- owid_address |>
  read_csv(
    col_types = cols(
      .default = col_double(),
      date = col_date(format = ""),
      iso_code = col_character(),
      location = col_character(),
      continent = col_character(),
      tests_units = col_character()
    ),
    progress = FALSE,
    show_col_types = FALSE
  )

covid_cases
```

```
## # A tibble: 429,435 x 67
##   iso_code continent location    date    total_cases new_cases new_cases_smoothed total_deaths n
##   <chr>    <chr>    <chr>    <date>         <dbl>     <dbl>         <dbl>         <dbl>
## 1 AFG      Asia      Afghanistan 2020-01-05         0         0             NA         0
## 2 AFG      Asia      Afghanistan 2020-01-06         0         0             NA         0
## 3 AFG      Asia      Afghanistan 2020-01-07         0         0             NA         0
## 4 AFG      Asia      Afghanistan 2020-01-08         0         0             NA         0
## 5 AFG      Asia      Afghanistan 2020-01-09         0         0             NA         0
## 6 AFG      Asia      Afghanistan 2020-01-10         0         0              0         0
## 7 AFG      Asia      Afghanistan 2020-01-11         0         0              0         0
## 8 AFG      Asia      Afghanistan 2020-01-12         0         0              0         0
## 9 AFG      Asia      Afghanistan 2020-01-13         0         0              0         0
## 10 AFG     Asia      Afghanistan 2020-01-14         0         0              0         0
## # i 429,425 more rows
## # i 57 more variables: total_cases_per_million <dbl>, new_cases_per_million <dbl>, new_cases_smoothed_per_million <dbl>,
## #   total_deaths_per_million <dbl>, new_deaths_per_million <dbl>, new_deaths_smoothed_per_million <dbl>,
## #   icu_patients <dbl>, icu_patients_per_million <dbl>, hosp_patients <dbl>, hosp_patients_per_million <dbl>,
## #   weekly_icu_admissions_per_million <dbl>, weekly_hosp_admissions <dbl>, weekly_hosp_admissions_per_million <dbl>,
## #   new_tests <dbl>, total_tests_per_thousand <dbl>, new_tests_per_thousand <dbl>, new_tests_smoothed_per_thousand <dbl>,
## #   new_tests_smoothed_per_thousand <dbl>, positive_rate <dbl>, tests_per_case <dbl>, tests_units <chr>
```

The data are collected daily, which leads to a lot of data! This is probably more than we need to compare to the other variables.

In the chunk below, aggregate the data to be total cases in 2020, by country. Your solution should only have columns `country`, `continent` and `total_cases`. (*Hint*: The `year()` function will extract the year from a date column.)

```
# YOUR CODE HERE
covid_2020 <- covid_cases |>
  filter(year(date) == 2020) |>
  group_by(location, continent) |>
  summarise(total_cases = max(total_cases, na.rm = TRUE)) |>
```

```
ungroup() |>
rename(country = location)
```

```
## Warning: There were 6 warnings in 'summarise()'.
## The first warning was:
## i In argument: 'total_cases = max(total_cases, na.rm = TRUE)'.
## i In group 66: 'location = "England"' 'continent = "Europe"'.
## Caused by warning in 'max()':
## ! no non-missing arguments to max; returning -Inf
## i Run 'dplyr::last_dplyr_warnings()' to see the 5 remaining warnings.

## 'summarise()' has grouped output by 'location'. You can override using the '.groups' argument.
```

```
covid_2020
```

```
## # A tibble: 252 x 3
##   country      continent    total_cases
##   <chr>        <chr>         <dbl>
## 1 Afghanistan Asia           51848
## 2 Africa      <NA>         2662452
## 3 Albania     Europe          55380
## 4 Algeria     Africa          97857
## 5 American Samoa Oceania           0
## 6 Andorra     Europe           7806
## 7 Angola      Africa          17149
## 8 Anguilla    North America     12
## 9 Antigua and Barbuda North America    155
## 10 Argentina  South America   1629908
## # i 242 more rows
```

```
. = ottr::check("tests/covid_2020.R")
```

```
## Warning in file(filename, "r"): cannot open file 'tests/covid_2020.R': No such file or directory
## Error in file(filename, "r"): cannot open the connection
```

Before we move on, we'll save covid_2020.parquet for use in Part 2.

```
# YOUR CODE HERE
write_parquet(covid_2020, "covid_2020.parquet")
```

```
. = ottr::check("tests/covid_2020_parquet.R")
```

```
## Warning in file(filename, "r"): cannot open file 'tests/covid_2020_parquet.R': No such file or direc
## Error in file(filename, "r"): cannot open the connection
```

As a very early exploration, let's compare climate awareness to the total number of covid cases in a country *per capita* in the year 2020.

In order to calculate per capita values, we need the population size. The easiest way that I can find is with the R package `wbstats`, which pulls updated statistics from the World Bank.

```
# DO NOT PUT install.packages() IN AN RMD
# Run install.packages("wbstats") in your console, not in this code chunk
library(wbstats)
pop_data <- wb_data("SP.POP.TOTL", start_date = 2020, end_date = 2020)
```

A quick check to see which countries are in one data set but not the other:

```
pop_countries <- pop_data |> pull(country)
aware_countries <- climate_awareness |> pull(country)

print("Countries in pop_data that aren't in climate_awareness:")
```

```
## [1] "Countries in pop_data that aren't in climate_awareness:"
```

```
setdiff(pop_countries, aware_countries)
```

```
## [1] "Aruba"
## [5] "American Samoa"
## [9] "Bahrain"
## [13] "Belize"
## [17] "Brunei Darussalam"
## [21] "China"
## [25] "Comoros"
## [29] "Curacao"
## [33] "Dominican Republic"
## [37] "Ethiopia"
## [41] "Gabon"
## [45] "Guinea"
## [49] "Grenada"
## [53] "Hong Kong SAR, China"
## [57] "Kazakhstan"
## [61] "Korea, Rep."
## [65] "Liechtenstein"
## [69] "Latvia"
## [73] "Moldova"
## [77] "North Macedonia"
## [81] "Montenegro"
## [85] "Mauritius"
## [89] "Nauru"
## [93] "Puerto Rico (US)"
## [97] "Russian Federation"
## [101] "Solomon Islands"
## [105] "Somalia, Fed. Rep."
## [109] "Slovak Republic"
## [113] "Seychelles"
## [117] "Togo"
## [121] "Tonga"
## [125] "Uganda"
## [129] "Venezuela, RB"
## [133] "Vanuatu"
## [137] "Zimbabwe"

"Afghanistan"
"Antigua and Barbuda"
"Bahamas, The"
"Bermuda"
"Bhutan"
"Cote d'Ivoire"
"Cabo Verde"
"Cayman Islands"
"Egypt, Arab Rep."
"Fiji"
"United Kingdom"
"Gambia, The"
"Greenland"
"Isle of Man"
"Kyrgyz Republic"
"Lao PDR"
"Sri Lanka"
"Macao SAR, China"
"Madagascar"
"Mali"
"Mongolia"
"Namibia"
"New Zealand"
"Korea, Dem. People's Rep."
"Rwanda"
"Sierra Leone"
"South Sudan"
"Slovenia"
"Syrian Arab Republic"
"Tajikistan"
"Trinidad and Tobago"
"Ukraine"
"British Virgin Islands"
"Samoa"

"Andorra"
"Burundi"
"Bosnia and Herzegovina"
"Bolivia"
"Central African Republic"
"Congo, Dem. Rep."
"Costa Rica"
"Djibouti"
"Eritrea"
"Faroe Islands"
"Georgia"
"Guinea-Bissau"
"Guam"
"Iran, Islamic Rep."
"Kiribati"
"Liberia"
"Lesotho"
"St. Martin (French part)"
"Maldives"
"Malta"
"Northern Mariana Islands"
"New Caledonia"
"Palau"
"West Bank and Gaza"
"Saudi Arabia"
"El Salvador"
"Sao Tome and Principe"
"Eswatini"
"Turks and Caicos Islands"
"Turkmenistan"
"Turkiye"
"United States"
"Virgin Islands (U.S.)"
"Yemen, Rep."
```

```
print("Countries in aware_countries that aren't in pop_data:")
```

```
## [1] "Countries in aware_countries that aren't in pop_data:"
```

```
setdiff(aware_countries, pop_countries)
```

```
## [1] "Asian.&.Pacific.Islands"      "Bolivia.(Plurinational.State.of)"  "Bosnia.and.Herzegovina"
## [4] "Burkina.Faso"                "Caribbean"                        "Congo.(Democratic.Rep"
## [7] "Costa.Rica"                  "Cote.d'Ivoire"                    "Dominican.Republic"
## [10] "Egypt"                       "El.Salvador"                      "Hong.Kong"
## [13] "Lao.People's.Democratic.Republic" "New.Zealand"                      "North.Macedonia"
## [16] "Puerto.Rico"                "Saudi.Arabia"                     "Slovakia"
## [19] "South.Africa"                "South.Korea"                      "Sri.Lanka"
## [22] "Sub-Saharan.Africa"          "Taiwan"                           "Trinidad.and.Tobago"
## [25] "Turkey"                     "United.Arab.Emirates"             "United.Kingdom"
## [28] "United.States"              "Vietnam"                           "Yemen"
```

Many of these countries are actually in both, but have a different name! Before we can join these two data sets together, we need the primary key (`country`) to match where possible.

- One pattern that I notice is that `aware_countries` has periods (.) instead of spaces. As a first step (to save us from manually entering 29 different country names twice), change all periods to spaces using `str_replace()`.
 - NOTE: The . is a special character in regex and needs to be escaped!

```
aware_fixed_country <- climate_awareness |>
  mutate(
    # Replace *all* *literal* periods with a space in the country column
    # YOUR CODE HERE
    country = str_replace_all(country, "\\.", " ")
  )

aware_fixed_country |> pull(country) |> unique()
```

```
## [1] "Albania"      "Algeria"      "Angola"
## [4] "Argentina"   "Armenia"      "Asian & Pacific Isl"
## [7] "Australia"   "Austria"      "Azerbaijan"
## [10] "Bangladesh"  "Belgium"      "Benin"
## [13] "Bolivia (Plurinational State of)" "Bosnia and Herzegovina" "Botswana"
## [16] "Brazil"      "Bulgaria"     "Burkina Faso"
## [19] "Cambodia"    "Cameroon"     "Canada"
## [22] "Caribbean"   "Chile"        "Colombia"
## [25] "Congo (Democratic Republic of the)" "Costa Rica"      "Cote d'Ivoire"
## [28] "Croatia"     "Cyprus"        "Czechia"
## [31] "Denmark"     "Dominican Republic" "Ecuador"
## [34] "Egypt"       "El Salvador"   "Finland"
## [37] "France"      "Germany"       "Ghana"
## [40] "Greece"      "Guatemala"     "Haiti"
## [43] "Honduras"    "Hong Kong"     "Hungary"
## [46] "India"       "Indonesia"     "Iraq"
```

```
## [49] "Ireland" "Israel" "Italy"
## [52] "Jamaica" "Japan" "Jordan"
## [55] "Kenya" "Kosovo" "Kuwait"
## [58] "Lao People's Democratic Republic" "Lebanon" "Libya"
## [61] "Lithuania" "Malawi" "Malaysia"
## [64] "Mexico" "Morocco" "Mozambique"
## [67] "Nepal" "Netherlands" "New Zealand"
## [70] "Nicaragua" "Nigeria" "North Macedonia"
## [73] "Norway" "Oman" "Pakistan"
## [76] "Panama" "Paraguay" "Peru"
## [79] "Philippines" "Poland" "Portugal"
## [82] "Puerto Rico" "Qatar" "Romania"
## [85] "Saudi Arabia" "Senegal" "Serbia"
## [88] "Singapore" "Slovakia" "South Africa"
## [91] "South Korea" "Spain" "Sri Lanka"
## [94] "Sub-Saharan Africa" "Sweden" "Switzerland"
## [97] "Taiwan" "Tanzania" "Thailand"
## [100] "Trinidad and Tobago" "Tunisia" "Turkey"
## [103] "United Arab Emirates" "United Kingdom" "United States"
## [106] "Uruguay" "Uzbekistan" "Vietnam"
## [109] "Yemen" "Zambia"
```

```
. = ottr::check("tests/aware_fixed_country.R")
```

```
## Warning in file(filename, "r"): cannot open file 'tests/aware_fixed_country.R': No such file or dire
```

```
## Error in file(filename, "r"): cannot open the connection
```

Now let's run the check again:

```
pop_countries <- pop_data |> pull(country)
aware_countries <- aware_fixed_country |> pull(country)

print("Countries in pop_data that aren't in climate_awareness:")
```

```
## [1] "Countries in pop_data that aren't in climate_awareness:"
```

```
setdiff(pop_countries, aware_countries) |> sort()
```

```
## [1] "Afghanistan" "American Samoa" "Andorra"
## [5] "Aruba" "Bahamas, The" "Bahrain"
## [9] "Belarus" "Belize" "Bermuda"
## [13] "Bolivia" "British Virgin Islands" "Brunei Darussalam"
## [17] "Cabo Verde" "Cayman Islands" "Central African Republic"
## [21] "Channel Islands" "China" "Comoros"
## [25] "Congo, Rep." "Cuba" "Curacao"
## [29] "Dominica" "Egypt, Arab Rep." "Equatorial Guinea"
## [33] "Estonia" "Eswatini" "Ethiopia"
## [37] "Fiji" "French Polynesia" "Gabon"
## [41] "Georgia" "Gibraltar" "Greenland"
## [45] "Guam" "Guinea" "Guinea-Bissau"
```

```
## [49] "Hong Kong SAR, China"      "Iceland"      "Iran, Islamic Rep."
## [53] "Kazakhstan"               "Kiribati"     "Korea, Dem. People's Rep."
## [57] "Kyrgyz Republic"         "Lao PDR"      "Latvia"
## [61] "Liberia"                  "Liechtenstein" "Luxembourg"
## [65] "Madagascar"             "Maldives"     "Mali"
## [69] "Marshall Islands"        "Mauritania"   "Mauritius"
## [73] "Moldova"                  "Monaco"       "Mongolia"
## [77] "Myanmar"                  "Namibia"      "Nauru"
## [81] "Niger"                    "North Macedonia" "Northern Mariana Islands"
## [85] "Papua New Guinea"         "Puerto Rico (US)" "Russian Federation"
## [89] "Samoa"                    "San Marino"   "Sao Tome and Principe"
## [93] "Sierra Leone"           "Sint Maarten (Dutch part)" "Slovak Republic"
## [97] "Solomon Islands"         "Somalia, Fed. Rep." "South Sudan"
## [101] "St. Lucia"               "St. Martin (French part)" "St. Vincent and the Grenadines"
## [105] "Suriname"                 "Syrian Arab Republic" "Tajikistan"
## [109] "Togo"                     "Tonga"        "Turkiye"
## [113] "Turks and Caicos Islands" "Tuvalu"       "Uganda"
## [117] "Vanuatu"                  "Venezuela, RB" "Viet Nam"
## [121] "West Bank and Gaza"      "Yemen, Rep."  "Zimbabwe"
```

```
print("Countries in aware_countries that aren't in pop_data:")
```

```
## [1] "Countries in aware_countries that aren't in pop_data:"
```

```
setdiff(aware_countries, pop_countries) |> sort()
```

```
## [1] "Asian & Pacific Islands"      "Bolivia (Plurinational State of)" "Caribbean"
## [4] "Congo (Democratic Republic of the)" "Egypt" "Hong Kong"
## [7] "Lao People's Democratic Republic" "North Macedonia" "Puerto Rico"
## [10] "Slovakia" "South Korea" "Sub-Saharan Africa"
## [13] "Taiwan" "Turkey" "Vietnam"
## [16] "Yemen"
```

That's slightly less than what we need to deal with! Overall, the `pop_data` has more countries in it than `aware_fixed_country`, so we'll just do our best to change the countries in `aware_fixed_country` to match `pop_data`. Note that there a lot of controversial politics involved in defining a “country” (esp. Taiwan, Hong Kong, West Bank and Gaza, etc.), and I'll avoid these as much as possible for now.

Use a `case_when()` statement to change the countries in `aware_fixed_country` where possible. Specifically, fix the following:

- “Bolivia (Plurinational State of)”, “Congo (Democratic Republic of the)”, “Egypt”, “Hong Kong”, “Lao People’s Democratic Republic”, “Slovakia”, “North Macedonia”, “South Korea” (be careful with this one!), “Turkey”, “Vietnam”, “Yemen”.

```
aware_fixed_country_match <- aware_fixed_country |>
  mutate(
    # Change country using a case_when statement
    # YOUR CODE HERE
    country = case_when(
      country == "Bolivia (Plurinational State of)" ~ "Bolivia",
      country == "Congo (Democratic Republic of the)" ~ "Congo, Dem. Rep.",
```

```

country == "Egypt" ~ "Egypt, Arab Rep.",
country == "Hong Kong" ~ "Hong Kong SAR, China",
country == "Lao People's Democratic Republic" ~ "Lao PDR",
country == "Slovakia" ~ "Slovak Republic",
country == "North Macedonia" ~ "North Macedonia",
country == "South Korea" ~ "Korea, Rep.",
country == "Turkey" ~ "Turkiye",
country == "Vietnam" ~ "Viet Nam",
country == "Yemen" ~ "Yemen, Rep.",
# country == "Puerto Rico" ~ "Puerto Rico (US)",
TRUE ~ country
)
)

aware_countries <- aware_fixed_country_match |> pull(country)
print("Countries in aware_countries that aren't in pop_data:")

```

```
## [1] "Countries in aware_countries that aren't in pop_data:"
```

```
setdiff(aware_countries, pop_countries) |> sort()
```

```
## [1] "Asian & Pacific Islands" "Caribbean" "Puerto Rico" "Sub-Saharan Africa"
```

```

pop_countries <- pop_data |> pull(country)
aware_countries <- aware_fixed_country_match |> pull(country)

setdiff(aware_countries, pop_countries) |> sort() |> length()

```

```
## [1] 5
```

```
. = ottr::check("tests/aware_fixed_country_match.R")
```

```
## Warning in file(filename, "r"): cannot open file 'tests/aware_fixed_country_match.R': No such file or directory
```

```
## Error in file(filename, "r"): cannot open the connection
```

The remaining countries are either broader geopolitical areas rather than countries, or Taiwan (which is apparently recognized as a country by Meta but not World Bank). For simplicity, we will simply move on without these countries/areas.

Now we can join! Use a join that only keeps countries that are in both data sets. The final data frame should have the columns `country`, `total_population`, `aware_no`, `aware_alittle`, `aware_moderate`, `aware_alot`, and `aware_refuse`.

```

# YOUR CODE HERE
aware_pop <- aware_fixed_country_match |>
  filter(country != "Puerto Rico (US)") |>
  inner_join(
    pop_data |>
      select(country, total_population = SP.POP.TOTL),

```

```

    by = "country"
  ) |>
  select(
    country,
    total_population,
    aware_no,
    aware_alittle,
    aware_moderate,
    aware_alot,
    aware_refuse
  )
aware_pop

```

```

## # A tibble: 105 x 7
##   country      total_population aware_no aware_alittle aware_moderate aware_alot aware_refuse
##   <chr>          <dbl>      <dbl>      <dbl>          <dbl>      <dbl>      <dbl>
## 1 Albania          2837849      9.99        43.0           33.1       10.6       3.25
## 2 Algeria          44042091     23.2        37.6           26.0       11.1       2.15
## 3 Angola           33451132     19.1        43.0           23.5       11.1       3.29
## 4 Argentina        45191965      8.25        37.4           43.8        9.88      0.696
## 5 Armenia           2961500      9.18        51.4           25.8       11.4       2.14
## 6 Australia        25649248      1.13        26.4           48.1       23.9      0.407
## 7 Austria           8916864      1.34        16.4           48.6       32.7      0.985
## 8 Azerbaijan        10093121      6.19        42.4           35.9       13.2       2.20
## 9 Bangladesh       166298024     24.5        36.2           16.2       18.9      4.22
## 10 Belgium          11538604      1.40        26.9           57.0       14.6     0.0961
## # i 95 more rows

```

```

. = ottr::check("tests/aware_pop.R")

```

```

## Warning in file(filename, "r"): cannot open file 'tests/aware_pop.R': No such file or directory

```

```

## Error in file(filename, "r"): cannot open the connection

```

Finally, let's make sure the country names in `covid_2020` match those in `aware_pop`. In the end, all countries in `covid_2020` should be in `aware_pop` (but `aware_pop` will have countries that are not present in `covid_2020_match_country`, which is unavoidable).

```

aware_countries <- aware_pop |> pull(country) |> unique()
covid_countries <- covid_2020 |> pull(country) |> unique()

setdiff(covid_countries, aware_countries)

```

```

##   [1] "Afghanistan"      "Africa"      "American Samoa"
##   [4] "Andorra"          "Anguilla"    "Antigua and Barbuda"
##   [7] "Aruba"            "Asia"        "Bahamas"
##  [10] "Bahrain"          "Barbados"    "Belarus"
##  [13] "Belize"           "Bermuda"     "Bhutan"
##  [16] "Bonaire Sint Eustatius and Saba" "British Virgin Islands" "Brunei"
##  [19] "Burundi"          "Cape Verde"  "Cayman Islands"
##  [22] "Central African Republic" "Chad"        "China"

```


## [25] "Comoros"	"Congo"	"Cook Islands"
## [28] "Cuba"	"Curacao"	"Democratic Republic of C"
## [31] "Djibouti"	"Dominica"	"East Timor"
## [34] "Egypt"	"England"	"Equatorial Guinea"
## [37] "Eritrea"	"Estonia"	"Eswatini"
## [40] "Ethiopia"	"Europe"	"European Union (27)"
## [43] "Falkland Islands"	"Faroe Islands"	"Fiji"
## [46] "French Guiana"	"French Polynesia"	"Gabon"
## [49] "Gambia"	"Georgia"	"Gibraltar"
## [52] "Greenland"	"Grenada"	"Guadeloupe"
## [55] "Guam"	"Guernsey"	"Guinea"
## [58] "Guinea-Bissau"	"Guyana"	"High-income countries"
## [61] "Hong Kong"	"Iceland"	"Iran"
## [64] "Isle of Man"	"Jersey"	"Kazakhstan"
## [67] "Kiribati"	"Kyrgyzstan"	"Laos"
## [70] "Latvia"	"Lesotho"	"Liberia"
## [73] "Liechtenstein"	"Low-income countries"	"Lower-middle-income cou"
## [76] "Luxembourg"	"Madagascar"	"Maldives"
## [79] "Mali"	"Malta"	"Marshall Islands"
## [82] "Martinique"	"Mauritania"	"Mauritius"
## [85] "Mayotte"	"Micronesia (country)"	"Moldova"
## [88] "Monaco"	"Mongolia"	"Montenegro"
## [91] "Montserrat"	"Myanmar"	"Namibia"
## [94] "Nauru"	"New Caledonia"	"Niger"
## [97] "Niue"	"North America"	"North Korea"
## [100] "Northern Ireland"	"Northern Mariana Islands"	"Oceania"
## [103] "Palau"	"Palestine"	"Papua New Guinea"
## [106] "Pitcairn"	"Puerto Rico"	"Reunion"
## [109] "Russia"	"Rwanda"	"Saint Barthelemy"
## [112] "Saint Helena"	"Saint Kitts and Nevis"	"Saint Lucia"
## [115] "Saint Martin (French part)"	"Saint Pierre and Miquelon"	"Saint Vincent and the G"
## [118] "Samoa"	"San Marino"	"Sao Tome and Principe"
## [121] "Scotland"	"Seychelles"	"Sierra Leone"
## [124] "Sint Maarten (Dutch part)"	"Slovakia"	"Slovenia"
## [127] "Solomon Islands"	"Somalia"	"South America"
## [130] "South Korea"	"South Sudan"	"Sudan"
## [133] "Suriname"	"Syria"	"Taiwan"
## [136] "Tajikistan"	"Togo"	"Tokelau"
## [139] "Tonga"	"Turkey"	"Turkmenistan"
## [142] "Turks and Caicos Islands"	"Tuvalu"	"Uganda"
## [145] "Ukraine"	"United States Virgin Islands"	"Upper-middle-income cou"
## [148] "Vanuatu"	"Vatican"	"Venezuela"
## [151] "Vietnam"	"Wales"	"Wallis and Futuna"
## [154] "World"	"Yemen"	"Zimbabwe"

```
setdiff(aware_countries, covid_countries)
```

## [1] "Congo, Dem. Rep."	"Egypt, Arab Rep."	"Hong Kong SAR, China"	"Lao PDR"	"Slo"
## [7] "Turkiye"	"Viet Nam"	"Yemen, Rep."		

```
# YOUR CODE HERE
```

```
covid_2020_match_country <- covid_2020 |>
```

```

filter(!is.na(continent)) |>      # remove regions & aggregates
select(country, continent, total_cases) |>
mutate(
  country = case_when(
    country == "Democratic Republic of Congo" ~ "Congo, Dem. Rep.",
    country == "Egypt" ~ "Egypt, Arab Rep.",
    country == "Hong Kong" ~ "Hong Kong SAR, China",
    country == "Laos" ~ "Lao PDR",
    country == "Slovakia" ~ "Slovak Republic",
    country == "South Korea" ~ "Korea, Rep.",
    country == "Turkey" ~ "Turkiye",
    country == "Vietnam" ~ "Viet Nam",
    country == "Yemen" ~ "Yemen, Rep.",
    TRUE ~ country
  )
)|>
semi_join(aware_pop, by = "country")
covid_2020_match_country

```

```

## # A tibble: 105 x 3
##   country    continent    total_cases
##   <chr>      <chr>          <dbl>
## 1 Albania    Europe           55380
## 2 Algeria    Africa           97857
## 3 Angola     Africa           17149
## 4 Argentina  South America    1629908
## 5 Armenia    Asia             157834
## 6 Australia  Oceania          28296
## 7 Austria    Europe           344732
## 8 Azerbaijan Asia             211764
## 9 Bangladesh Asia             509148
## 10 Belgium   Europe           638760
## # i 95 more rows

```

```
nrow(covid_2020_match_country)
```

```
## [1] 105
```

```
. = ottr::check("tests/covid_2020_match_country.R")
```

```
## Warning in file(filename, "r"): cannot open file 'tests/covid_2020_match_country.R': No such file or
```

```
## Error in file(filename, "r"): cannot open the connection
```

Okay, now take a quick breather. Grab a tea, go for a walk, rock out to some music, whatever self care you want. That was a lot of finicky stuff, but now we've got a data set that we can use!

Join `aware_pop` and `covid_2020_match_country`, again only keep countries that are in both. Create a new column called `covid_per_capita` based on the total cases divided by total population.

```
# YOUR CODE HERE
```

```
aware_covid <- aware_pop |>
  inner_join(covid_2020_match_country, by = "country") |>
  mutate(covid_per_capita = total_cases / total_population)
aware_covid
```

```
## # A tibble: 105 x 10
```

```
##   country      total_population aware_no aware_alittle aware_moderate aware_alot aware_refuse continent
##   <chr>          <dbl>      <dbl>      <dbl>      <dbl>      <dbl>      <dbl> <chr>
## 1 Albania        2837849         9.99        43.0        33.1        10.6         3.25 Europe
## 2 Algeria        44042091        23.2         37.6        26.0        11.1         2.15 Africa
## 3 Angola         33451132        19.1         43.0        23.5        11.1         3.29 Africa
## 4 Argentina      45191965         8.25         37.4        43.8         9.88      0.696 South A
## 5 Armenia        2961500         9.18         51.4        25.8        11.4         2.14 Asia
## 6 Australia      25649248         1.13         26.4        48.1        23.9         0.407 Oceania
## 7 Austria        8916864         1.34         16.4        48.6        32.7         0.985 Europe
## 8 Azerbaijan     10093121         6.19         42.4        35.9        13.2         2.20 Asia
## 9 Bangladesh     166298024        24.5         36.2        16.2        18.9         4.22 Asia
## 10 Belgium       11538604         1.40         26.9        57.0        14.6         0.0961 Europe
## # i 95 more rows
```

```
nrow(aware_covid)
```

```
## [1] 105
```

```
. = ottr::check("tests/aware_covid.R")
```

```
## Warning in file(filename, "r"): cannot open file 'tests/aware_covid.R': No such file or directory
```

```
## Error in file(filename, "r"): cannot open the connection
```

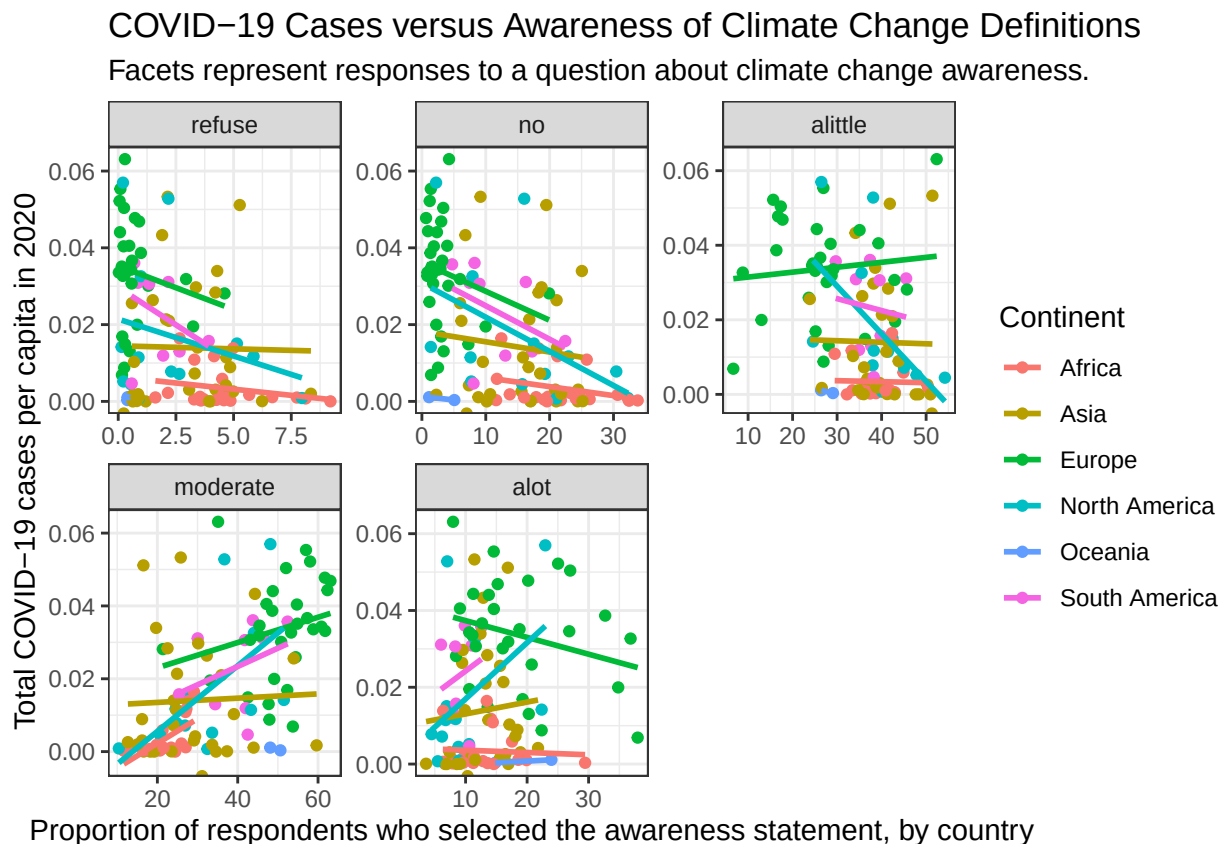
With all that done, I'll do the last few steps to make an interesting plot. Fortunately, the `covid_data` came with the classification of countries into continents, so we can add plenty of info to this plot!

```
aware_covid |>
  pivot_longer(
    cols = starts_with("aware_"),
    names_prefix = "aware_",
    names_to = "awareness",
    values_to = "value"
  ) |>
  mutate(
    awareness = ordered(awareness,
      levels = c("refuse", "no", "alittle", "moderate", "alot")
    )
  ) |>
  ggplot() +
  aes(x = value, y = covid_per_capita, colour = continent) + #it supposed to be covid_per_capita but
  geom_point() +
  geom_smooth(se = FALSE, method = "lm", formula = y ~ x) +
  facet_wrap(~ awareness, scales = "free") +
```

```
labs(
  x = "Proportion of respondents who selected the awareness statement, by country",
  y = "Total COVID-19 cases per capita in 2020",
  title = "COVID-19 Cases versus Awareness of Climate Change Definitions",
  subtitle = "Facets represent responses to a question about climate change awareness.",
  colour = "Continent"
)
```

```
## Warning: Removed 7 rows containing non-finite outside the scale range (‘stat_smooth()’).
```

```
## Warning: Removed 2 rows containing missing values or values outside the scale range (‘geom_point()’).
```



Some interesting patterns:

- It looks like the “refused” group are pretty similar to the “no” group. It seems safe to say that the people who refuse to answer specific questions on a climate survey are not aware of the actual definitions being used.
- For the Americas, it looks like being more aware of climate change is associated with an *increase* in COVID-19 cases.
- For other continents, there doesn’t appear to be any strong patterns.

There are several explanations for these patterns, such as awareness being linked to education and/or travel volume in first-world/developed countries, or perhaps it’s just a consequence of how we define a “continent” (could you list the 18 countries in North America?). Most likely some of the results are due to the differences in the way countries track and report cases of COVID-19.

3 World Happiness Report Score (from 2023 report)

Finally, we have data from the World Happiness Report, downloaded from here and distributed with this assignment. The “Statistical Appendix” includes the description of each column, especially what the question was.

For the most part, the “Ladder score” column is the most important - this is the WHR’s main question that gets at how “happy” the people of a country are. You may choose to explore other options though - perhaps people’s perceptions of corruption affect their response to climate issues?

To make the data more convenient for Part 2, perform the following operations. The first ones are easy, the last one is tricky!

- Use `janitor` to clean the names.
- Rename `Country_name` to `country`.
- Remove any rows for which `ladder_score` is NA.
- The data contain information from multiple years, and these years are not consistent for each country. For Part 2, these data will mainly be used to compare countries (we don’t need the happiness over time). Filter the data so that we only have the *most recent* observation of the relevant values. (In my solution, I use the `slice()` function as part of a very short pipeline.)

```
# First, let's see the structure of the data
happiness <- read.xlsx("DataForTable2.1.xlsx") |>
  clean_names() |>
  rename(country = country_name) |>
  filter(!is.na(ladder_score)) |>
  # Use arrange + distinct to get the most recent year
  arrange(country, desc(year)) |>
  distinct(country, .keep_all = TRUE) |>
  ungroup()
happiness
```

##	year	rank	country	ladder_score	lower_whisker	upper_whisker	explained_by_log_gdp
## 1	2024	147	Afghanistan	1.3640	1.301000	1.427000	
## 2	2024	89	Albania	5.4110	5.284000	5.539000	
## 3	2024	84	Algeria	5.5710	5.469000	5.674000	
## 4	2017	142	Angola	3.7950	NA	NA	
## 5	2024	42	Argentina	6.3970	6.302000	6.492000	
## 6	2024	87	Armenia	5.4940	5.391000	5.596000	
## 7	2024	11	Australia	6.9740	6.890000	7.057000	
## 8	2024	17	Austria	6.8100	6.728000	6.891000	
## 9	2024	106	Azerbaijan	4.8750	4.780000	4.969000	
## 10	2024	59	Bahrain	6.0300	5.891000	6.170000	
## 11	2024	134	Bangladesh	3.8510	3.753000	3.949000	
## 12	2021	65	Belarus	5.8210	5.693000	5.950000	
## 13	2024	14	Belgium	6.9100	6.845000	6.974000	
## 14	2024	25	Belize	6.7110	6.432000	6.989000	
## 15	2024	121	Benin	4.3570	4.222000	4.492000	
## 16	2018	95	Bhutan	5.0820	NA	NA	
## 17	2024	74	Bolivia	5.8680	5.758000	5.977000	
## 18	2024	56	Bosnia and Herzegovina	6.1360	6.039000	6.234000	
## 19	2024	142	Botswana	3.4380	3.301000	3.576000	
## 20	2024	36	Brazil	6.4940	6.394000	6.593000	
## 21	2024	85	Bulgaria	5.5540	5.460000	5.647000	

##	22	2024	120	Burkina Faso	4.3830	4.260000	4.506000
##	23	2020	141	Burundi	3.7750	3.565000	3.985000
##	24	2024	124	Cambodia	4.3410	4.237000	4.446000
##	25	2024	104	Cameroon	4.8870	4.767000	5.007000
##	26	2024	18	Canada	6.8030	6.718000	6.888000
##	27	2019	149	Central African Republic	3.4759	3.250141	3.701658
##	28	2024	119	Chad	4.3840	4.253000	4.515000
##	29	2024	45	Chile	6.3610	6.273000	6.449000
##	30	2024	68	China	5.9210	5.819000	6.023000
##	31	2024	61	Colombia	6.0040	5.897000	6.112000
##	32	2024	139	Comoros	3.7540	3.590000	3.919000
##	33	2024	100	Congo	5.0300	4.855000	5.205000
##	34	2024	6	Costa Rica	7.2740	7.187000	7.361000
##	35	2024	72	Croatia	5.8700	5.777000	5.963000
##	36	2011	69	Cuba	5.4180	NA	NA
##	37	2024	67	Cyprus	5.9420	5.827000	6.058000
##	38	2024	20	Czechia	6.7750	6.696000	6.854000
##	39	2024	98	Côte d'Ivoire	5.1020	4.967000	5.238000
##	40	2024	141	DR Congo	3.4690	3.331000	3.608000
##	41	2024	2	Denmark	7.5210	7.431000	7.611000
##	42	2014	126	Djibouti	4.3690	NA	NA
##	43	2024	76	Dominican Republic	5.8460	5.705000	5.988000
##	44	2024	62	Ecuador	5.9650	5.848000	6.083000
##	45	2024	135	Egypt	3.8170	3.725000	3.909000
##	46	2024	37	El Salvador	6.4920	6.365000	6.619000
##	47	2024	39	Estonia	6.4170	6.337000	6.497000
##	48	2024	137	Eswatini	3.7740	3.654000	3.893000
##	49	2024	132	Ethiopia	3.8980	3.786000	4.011000
##	50	2024	1	Finland	7.7360	7.662000	7.810000
##	51	2024	33	France	6.5930	6.517000	6.668000
##	52	2024	97	Gabon	5.1200	4.985000	5.256000
##	53	2024	117	Gambia	4.4230	4.315000	4.530000
##	54	2024	91	Georgia	5.4000	5.304000	5.496000
##	55	2024	22	Germany	6.7530	6.652000	6.854000
##	56	2024	125	Ghana	4.3400	4.235000	4.444000
##	57	2024	81	Greece	5.7760	5.677000	5.875000
##	58	2024	44	Guatemala	6.3620	6.222000	6.503000
##	59	2024	102	Guinea	4.9290	4.787000	5.070000
##	60	2011	45	Guyana	5.9930	NA	NA
##	61	2020	144	Haiti	3.6150	3.276000	3.953000
##	62	2024	63	Honduras	5.9640	5.821000	6.108000
##	63	2024	88	Hong Kong SAR of China	5.4910	5.417000	5.566000
##	64	2024	69	Hungary	5.9150	5.809000	6.020000
##	65	2024	3	Iceland	7.5150	7.425000	7.606000
##	66	2024	118	India	4.3890	4.320000	4.457000
##	67	2024	83	Indonesia	5.6170	5.516000	5.717000
##	68	2024	99	Iran	5.0930	4.994000	5.193000
##	69	2024	101	Iraq	4.9760	4.866000	5.085000
##	70	2024	15	Ireland	6.8890	6.801000	6.977000
##	71	2024	8	Israel	7.2340	7.168000	7.301000
##	72	2024	40	Italy	6.4150	6.309000	6.521000
##	73	2024	73	Jamaica	5.8700	5.563000	6.178000
##	74	2024	55	Japan	6.1470	6.063000	6.230000
##	75	2024	128	Jordan	4.3100	4.193000	4.426000

##	76	2024	43	Kazakhstan	6.3780	6.294000	6.462000	
##				explained_by_healthy_life_expectancy	explained_by_freedom_to_make_life_choices	explained_by_gener		
##	1			0.155		0.0000000		0.07
##	2			0.696		0.8410000		0.10
##	3			0.782		0.5220000		0.05
##	4			NA		NA		
##	5			0.579		0.8240000		0.04
##	6			0.828		0.7860000		0.03
##	7			0.841		0.8570000		0.16
##	8			0.814		0.8560000		0.17
##	9			0.707		0.7850000		0.07
##	10			0.609		0.9070000		0.14
##	11			0.592		0.9200000		0.11
##	12			0.629		0.3420000		0.04
##	13			0.852		0.8950000		0.11
##	14			0.525		1.0180000		0.07
##	15			0.357		0.6790000		0.09
##	16			NA		NA		
##	17			0.346		0.8170000		0.06
##	18			0.616		0.7490000		0.17
##	19			0.094		0.6970000		0.01
##	20			0.480		0.8300000		0.08
##	21			0.463		0.7080000		0.05
##	22			0.296		0.6710000		0.14
##	23			0.155		0.2980000		0.17
##	24			0.481		0.9880000		0.13
##	25			0.302		0.6620000		0.09
##	26			0.832		0.8130000		0.16
##	27			0.000		0.2928145		0.25
##	28			0.242		0.4770000		0.17
##	29			0.711		0.7790000		0.09
##	30			0.789		0.7640000		0.07
##	31			0.587		0.7770000		0.03
##	32			0.428		0.2600000		0.11
##	33			0.336		0.6260000		0.08
##	34			0.680		0.9480000		0.06
##	35			0.669		0.4780000		0.02
##	36			NA		NA		
##	37			0.840		0.6120000		0.10
##	38			0.658		0.9160000		0.11
##	39			0.206		0.6610000		0.10
##	40			0.278		0.6000000		0.15
##	41			0.820		0.9550000		0.15
##	42			NA		NA		
##	43			0.615		0.8660000		0.06
##	44			0.702		0.7120000		0.05
##	45			0.473		0.5930000		0.01
##	46			0.559		0.9550000		0.05
##	47			0.642		0.9020000		0.12
##	48			0.038		0.4280000		0.05
##	49			0.416		0.5400000		0.18
##	50			0.824		0.9860000		0.11
##	51			0.832		0.7430000		0.09
##	52			0.387		0.6510000		0.04

## 53	0.348	0.6110000	0.255
## 54	0.482	0.8440000	0.000
## 55	0.776	0.8580000	0.140
## 56	0.438	0.7710000	0.130
## 57	0.725	0.3960000	0.000
## 58	0.376	0.8820000	0.070
## 59	0.306	0.6760000	0.150
## 60	NA	NA	
## 61	0.227	0.2570000	0.460
## 62	0.436	0.8620000	0.130
## 63	0.948	0.6740000	0.110
## 64	0.587	0.6810000	0.090
## 65	0.873	0.9710000	0.200
## 66	0.316	0.9140000	0.140
## 67	0.454	0.9180000	0.320
## 68	0.609	0.4560000	0.200
## 69	0.589	0.5030000	0.100
## 70	0.823	0.9060000	0.170
## 71	0.824	0.7400000	0.140
## 72	0.852	0.6410000	0.080
## 73	0.504	0.8690000	0.070
## 74	0.921	0.7460000	0.010
## 75	0.573	0.7000000	0.050
## 76	0.504	0.8780000	0.110
##	explained_by_perceptions_of_corruption dystopia_residual		
## 1	0.13500000	0.348000	
## 2	0.05700000	1.220000	
## 3	0.21900000	1.313000	
## 4	NA	NA	
## 5	0.08300000	1.736000	
## 6	0.15800000	0.950000	
## 7	0.28500000	1.413000	
## 8	0.28000000	1.297000	
## 9	0.17900000	0.670000	
## 10	0.16700000	1.057000	
## 11	0.13200000	0.695000	
## 12	0.28200000	1.811000	
## 13	0.30600000	1.309000	
## 14	0.08200000	2.218000	
## 15	0.23800000	1.909000	
## 16	NA	NA	
## 17	0.05200000	2.057000	
## 18	0.00900000	1.635000	
## 19	0.08300000	0.012000	
## 20	0.12700000	2.057000	
## 21	0.00000000	1.029000	
## 22	0.21800000	1.504000	
## 23	0.21200000	2.876000	
## 24	0.07800000	0.393000	
## 25	0.06000000	1.834000	
## 26	0.32300000	1.271000	
## 27	0.02826472	2.860198	
## 28	0.11100000	1.884000	
## 29	0.08500000	1.579000	


```
## 30      0.15400000      1.382000
## 31      0.05100000      1.658000
## 32      0.16400000      1.450000
## 33      0.12600000      2.040000
## 34      0.11800000      2.369000
## 35      0.06500000      1.319000
## 36      NA      NA
## 37      0.04100000      1.273000
## 38      0.08400000      1.591000
## 39      0.16200000      2.176000
## 40      0.06100000      0.906000
## 41      0.48800000      1.535000
## 42      NA      NA
## 43      0.18600000      1.160000
## 44      0.06600000      1.808000
## 45      0.10300000      0.210000
## 46      0.26400000      2.211000
## 47      0.37800000      1.001000
## 48      0.13000000      0.790000
## 49      0.10000000      0.852000
## 50      0.50200000      1.782000
## 51      0.26400000      1.386000
## 52      0.08600000      1.343000
## 53      0.04500000      1.428000
## 54      0.17200000      1.182000
## 55      0.35100000      1.210000
## 56      0.02500000      0.904000
## 57      0.06300000      1.543000
## 58      0.08500000      2.365000
## 59      0.09800000      2.027000
## 60      NA      NA
## 61      0.13900000      2.060000
## 62      0.07900000      2.229000
## 63      0.38600000      0.168000
## 64      0.05700000      1.111000
## 65      0.17300000      1.659000
## 66      0.12000000      0.889000
## 67      0.04400000      1.218000
## 68      0.12200000      1.007000
## 69      0.06900000      1.178000
## 70      0.38200000      0.952000
## 71      0.19300000      1.895000
## 72      0.07100000      1.517000
## 73      0.01900000      1.676000
## 74      0.20000000      1.038000
## 75      0.16400000      0.461000
## 76      0.11800000      1.492000
## [ reached 'max' / getOption("max.print") -- omitted 92 rows ]
```

```
. = ottr::check("tests/happiness.R")
```

```
## Warning in file(filename, "r"): cannot open file 'tests/happiness.R': No such file or directory
```

```
## Error in file(filename, "r"): cannot open the connection
```

And, finally, save the file:

```
write_parquet(happiness, "happiness.parquet")
```

```
# END ASSIGNMENT
```

List your group members, including their student numbers, here: - Sophia Thai (#169112540) - Kaya Darling(#169112257)

4 Data Description

4.1 Climate Change Awareness Data

```
## # A tibble: 110 x 7
##   country          aware_no aware_alittle aware_moderate aware_alot aware_refuse aware_base
##   <chr>          <dbl>      <dbl>          <dbl>      <dbl>      <dbl>      <dbl>
## 1 Albania            9.99        43.0            33.1        10.6         3.25        329
## 2 Algeria           23.2         37.6            26.0        11.1         2.15       1066
## 3 Angola            19.1         43.0            23.5        11.1         3.29        727
## 4 Argentina          8.25         37.4            43.8         9.88         0.696      1257
## 5 Armenia            9.18         51.4            25.8        11.4         2.14        306
## 6 Asian & Pacific Islands 13.3         40.1            24.8        19.5         2.26       1916
## 7 Australia          1.13         26.4            48.1        23.9         0.407      1012
## 8 Austria            1.34         16.4            48.6        32.7         0.985      1120
## 9 Azerbaijan          6.19         42.4            35.9        13.2         2.20        541
## 10 Bangladesh        24.5         36.2            16.2        18.9         4.22        910
## # i 100 more rows
```

The data come from HDX (Humanitarian Data Exchange) and describes the responses to a survey in 2022 from the Meta Climate Change Opinion Survey. This survey asked people in many different countries their level of awareness on climate change and recorded their responses ¹. The people doing the survey had the option to choose between different statements that represented their awareness of climate change. For example, if they were not aware at all the participant would select “I have never heard of it”.

In order to clean the data, we first adjusted so that all of the periods in between names were removed and replaced with spaces. Then, we adjusted the country names that were called something different from their name in the other data sets to match the other data sets country names. This was to ensure that in joining everything goes smoothly.

4.2 Covid 2020 Cases Data

```
## # A tibble: 104 x 9
##   country          aware_no aware_alittle aware_moderate aware_alot aware_refuse aware_base continent
##   <chr>          <dbl>      <dbl>          <dbl>      <dbl>      <dbl>      <dbl> <chr>
## 1 Albania            9.99        43.0            33.1        10.6         3.25        329 Europe
## 2 Algeria           23.2         37.6            26.0        11.1         2.15       1066 Africa
## 3 Angola            19.1         43.0            23.5        11.1         3.29        727 Africa
## 4 Argentina          8.25         37.4            43.8         9.88         0.696      1257 South America
## 5 Armenia            9.18         51.4            25.8        11.4         2.14        306 Asia
```

¹Meta. (2022). Meta Climate Change Opinion Survey 2022 [Data set]. Humanitarian Data Exchange (HDX). <https://data.humdata.org/dataset/meta-climate-change-opinion-survey>

```
## 6 Australia      1.13      26.4      48.1      23.9      0.407      1012 Oceania
## 7 Austria        1.34      16.4      48.6      32.7      0.985      1120 Europe
## 8 Azerbaijan     6.19      42.4      35.9      13.2      2.20       541 Asia
## 9 Bangladesh    24.5      36.2      16.2      18.9      4.22       910 Asia
## 10 Belgium       1.40      26.9      57.0      14.6      0.0961     1201 Europe
## # i 94 more rows
```

The data come from the Our World in Data COVID_19 database and detail the number of covid cases in each country in 2020, recorded daily ².

In order to clean the data, we adjusted some of the names to match the climate awareness data by changing some of the names of certain countries so that they all match.

4.3 World Happiness Report Data

```
## # A tibble: 97 x 10
##   country      aware_no aware_alittle aware_moderate aware_alot aware_refuse aware_base continent
##   <chr>         <dbl>         <dbl>         <dbl>         <dbl>         <dbl>         <dbl> <chr>
## 1 Albania       9.99          43.0          33.1          10.6          3.25          329 Europe
## 2 Algeria      23.2          37.6          26.0          11.1          2.15         1066 Africa
## 3 Angola       19.1          43.0          23.5          11.1          3.29          727 Africa
## 4 Argentina     8.25          37.4          43.8           9.88          0.696        1257 South America
## 5 Armenia       9.18          51.4          25.8          11.4          2.14          306 Asia
## 6 Australia     1.13          26.4          48.1          23.9          0.407        1012 Oceania
## 7 Austria       1.34          16.4          48.6          32.7          0.985        1120 Europe
## 8 Azerbaijan     6.19          42.4          35.9          13.2          2.20          541 Asia
## 9 Bangladesh    24.5          36.2          16.2          18.9          4.22          910 Asia
## 10 Belgium      1.40          26.9          57.0          14.6          0.0961       1201 Europe
## # i 87 more rows
```

This data comes from the World Happiness Report from 2023 and detail how people responded to a question about the “Life Ladder” ³ about how they rate their overall life satisfaction from 0 to 10. The data has the overall well being of a country depending on how participants from these countries rated their life on the ladder.

In order to clean the data, we didn’t have to do any cleaning, as it was already clean from part 1!

4.4 Combining the Data

We first joined climate awareness data to covid data by selecting only columns “country”, “continent”, and `total_cases`. Then, we used `inner_join` to join those columns of the covid 2020 data with the climate awareness data.

For combining happiness data with climate awareness and covid data, we joined this data to the data set `climate_covid`, which was made from joining the previous two data sets. We used `inner_join()` to join the columns “country” and “ladder_score” from the happiness data to the climate awareness and covid data.

²Ritchie, H., Mathieu, E., Rod s-Guirao, L., Appel, C., Giattino, C., Ortiz-Ospina, E., Hasell, J., Macdonald, B., Beltekian, D., & Roser, M. (2020). Coronavirus pandemic (COVID-19) [Data set]. Our World in Data. <https://ourworldindata.org/coronavirus>

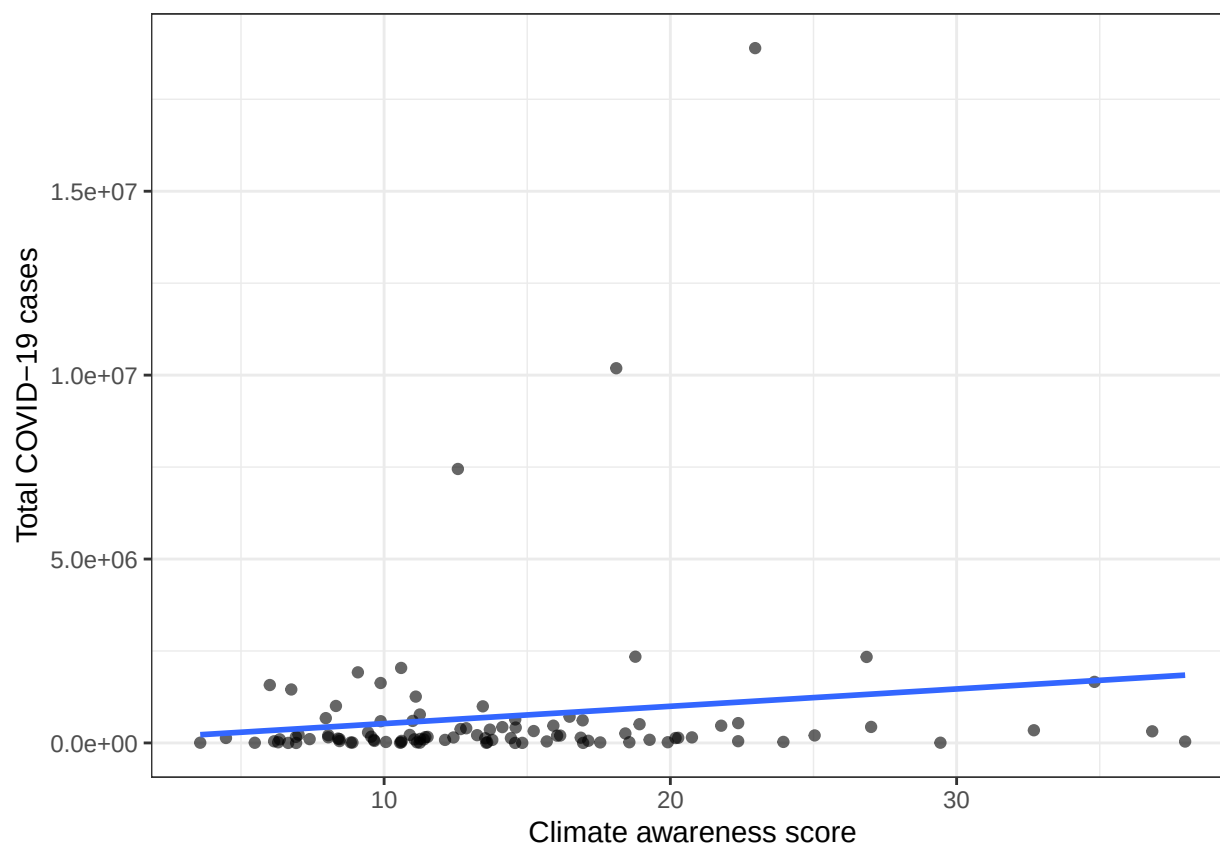
³Helliwell, J. F., Layard, R., Sachs, J. D., & De Neve, J.-E. (2023). World Happiness Report 2023 [Data set]. Sustainable Development Solutions Network. <https://worldhappiness.report>

5 Exploratory Data Analysis

To achieve our goals, we explored the data by...

We explored many aspects of the data, but will demonstrate three. These are countries with higher covid cases there are more people reporting high climate awareness, countries with higher climate awareness are happier, and countries with higher covid cases are also happier

The first aspect that we found interesting is shown in this scatterplot. The insight should be specific to the data shown, not a general statement beyond the data (leave that for the conclusion).



We observe a positive relationship between the percentage of respondents who reported “I know a lot about climate change” and the total number of COVID-19 cases ⁴⁵. This pattern does not imply that climate awareness causes higher case counts; rather, it reflects broader national characteristics. Countries with stronger education systems and greater access to information tend to show higher levels of climate awareness. These same countries also typically have higher population densities, more international travel, and more comprehensive COVID-19 testing and reporting infrastructure. Together, these factors produce the positive trend observed in the data.

This insight is supported by the summary statistics in table @ref(tab:summary_stats)

metric	correlation
Correlation	0.144

The next insight that we found is shown in 1.

⁴Meta. (2022). Meta Climate Change Opinion Survey 2022 [Data set]. Humanitarian Data Exchange (HDX). <https://data.humdata.org/dataset/meta-climate-change-opinion-survey>

⁵Ritchie, H., Mathieu, E., Rod  s-Guirao, L., Appel, C., Giattino, C., Ortiz-Ospina, E., Hasell, J., Macdonald, B., Beltekian, D., & Roser, M. (2020). Coronavirus pandemic (COVID-19) [Data set]. Our World in Data. <https://ourworldindata.org/coronavirus>

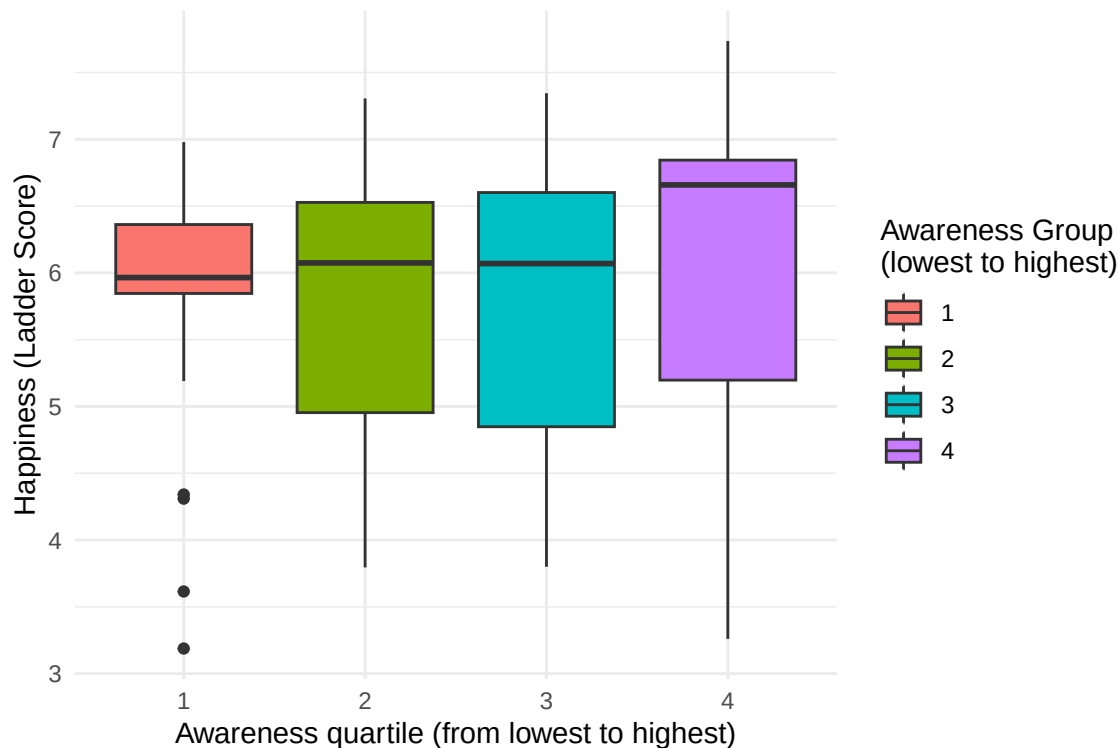


Figure 1: Happiness scores by climate awareness quartile

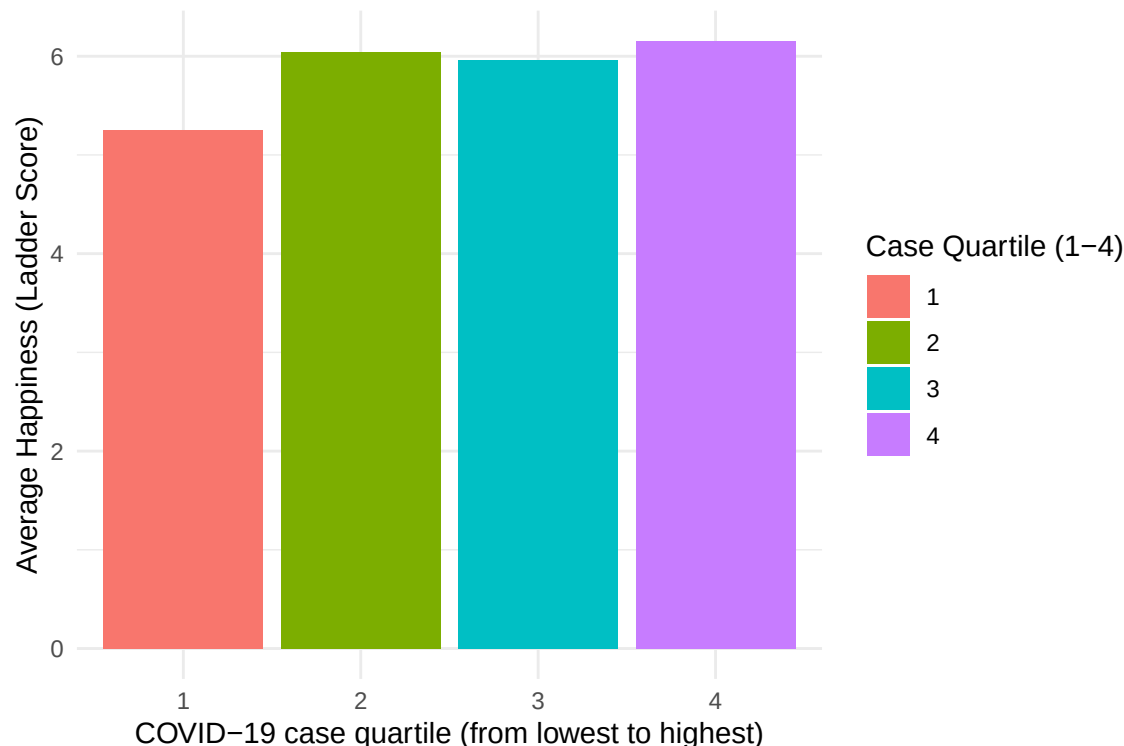
Countries with greater climate awareness tend to have higher reported happiness scores, based on data from the Meta Climate Change Survey and the World Happiness Report ⁶⁷.

This boxplot shows a clear upward trend, indicating that countries with greater climate awareness tend to have higher reported happiness scores. This pattern likely reflects broader structural differences rather than a direct causal relationship. Countries with higher climate awareness often also have better access to education, higher GDP, more stable institutions, and stronger scientific and media communication systems. These underlying factors contribute both to increased awareness of climate issues and to higher Life Ladder scores, which helps explain the positive association observed in the data.

Finally, ?? shows ...

⁶Meta. (2022). Meta Climate Change Opinion Survey 2022 [Data set]. Humanitarian Data Exchange (HDX). <https://data.humdata.org/dataset/meta-climate-change-opinion-survey>

⁷Ritchie, H., Mathieu, E., Rod s-Guirao, L., Appel, C., Giattino, C., Ortiz-Ospina, E., Hasell, J., Macdonald, B., Beltekian, D., & Roser, M. (2020). Coronavirus pandemic (COVID-19) [Data set]. Our World in Data. <https://ourworldindata.org/coronavirus>



countries in the highest COVID-case quartile also report the highest average happiness, which reflects patterns seen in COVID-19 reporting and Life Ladder data ⁸⁹.

Surprisingly, countries in the highest COVID-case quartile also report the highest average happiness. This counterintuitive pattern can be explained by the fact that regions such as Western Europe, North America, Australia, and parts of East Asia both experienced high reported case counts and consistently rank among the highest on the Life Ladder. In contrast, many countries with limited testing capacity reported fewer cases but also tend to have lower happiness scores. As a result, the association between higher case counts and higher happiness is indirect and largely reflects differences in economic development and infrastructure rather than any meaningful effect of COVID-19 itself.

6 Conclusion and Future Work

Overall, we found three consistent patterns across the disparate datasets. First, countries with more reported COVID-19 cases tend to show higher levels of climate awareness. This likely reflects stronger education systems, more robust scientific communication, and better reporting capacity in more developed regions. Second, higher climate awareness is associated with higher national happiness, suggesting that climate literacy may be part of a broader set of educational, social, and institutional advantages that contribute to well-being. Third, countries with the highest numbers of COVID-19 cases also rank among the happiest. Rather than indicating any causal relationship, this pattern appears to be driven by differences in economic development: wealthier countries both test more extensively and consistently report higher Life Ladder scores. Together, these findings suggest that global indicators such as public health outcomes, climate awareness,

⁸Ritchie, H., Mathieu, E., Rod  s-Guirao, L., Appel, C., Giattino, C., Ortiz-Ospina, E., Hasell, J., Macdonald, B., Beltekian, D., & Roser, M. (2020). Coronavirus pandemic (COVID-19) [Data set]. Our World in Data. <https://ourworldindata.org/coronavirus>

⁹Helliwell, J. F., Layard, R., Sachs, J. D., & De Neve, J.-E. (2023). World Happiness Report 2023 [Data set]. Sustainable Development Solutions Network. <https://worldhappiness.report>

and subjective well-being are interconnected through underlying socioeconomic structures ¹⁰¹¹.

For future work, several extensions could strengthen this analysis. Incorporating variables such as GDP, literacy rates, or education indexes would help control for differences in development and allow for a clearer interpretation of the relationships observed ¹²¹³. Multiple regression models could be used to distinguish direct from indirect effects across variables. Further work might also explore whether higher climate awareness predicts stronger support for climate policy or behavioral change at the national level. Additionally, explicitly including population size—through metrics like COVID-19 cases per capita—could improve the comparability of countries with very different populations.

This analysis also has several limitations. Measures of climate awareness rely on self-reported survey responses, which may be influenced by sampling quality and respondent honesty. Data inconsistencies, such as mismatched country names, require manual cleaning and may result in the exclusion of some countries. COVID-19 case data are affected by differences in testing capacity, meaning that wealthier countries often appear to have more cases simply because they tested more extensively. Most importantly, the relationships identified are correlational rather than causal; they reflect deeper structural factors rather than direct effects between the variables. Finally, sample sizes vary considerably across countries, which may affect the stability of group comparisons.

Despite these limitations, our findings highlight meaningful and interpretable patterns across global climate literacy, public health outcomes, and well-being. The integration of disparate datasets provides valuable insight into how social, economic, and institutional factors shape international trends in awareness, happiness, and reported health data.

6.1 Overview

This is **Part 2** of the project, in which you'll work with the data that you cleaned in Part 1!

There are several data sets that you cleaned in Part 1. Using at least **three** of these data sets, find something interesting!

Possible ideas:

- Is the intensity of hurricanes related to the sea ice extent?
- Is being aware of climate change associated with being less happy?

Note: You do not have to have fully completed Part 1 in order to do this part. You will need to use the cleaned version of *three* of the data in the other assignments, but you can focus on the important data first. In addition, using the `covid_2020` only for the continents information or using the `pop_data` do *not* count towards the three data sets.

Once you're a member of a group, you will see "Assignment 3 Report" in the Dropbox tool on MyLS. If you do not see this tool, it likely means that you are not yet in a group. Even if you're going solo, you must be a member of a group.

Collaborating across 2 files with 5 people can be hard. Here are some suggestions for collaboration with group members:

- Store the Rmds in a shared OneDrive folder, along with the data and the `tests` folder (everything that comes in the zip file).

¹⁰The World Bank. (2023). Country and lending groups [Data set]. <https://datahelpdesk.worldbank.org/knowledgebase/articles/906519>

¹¹United Nations Development Programme. (2023). Human Development Report 2023 [Data set]. <https://hdr.undp.org>

¹²The World Bank. (2023). Country and lending groups [Data set]. <https://datahelpdesk.worldbank.org/knowledgebase/articles/906519>

¹³United Nations Development Programme. (2023). Human Development Report 2023 [Data set]. <https://hdr.undp.org>

- Create an account on `posit.cloud`, and share the login information amongst yourselves. This makes it easy to avoid issues with package installation and working directory issues, and can be accessed from any device. Uploading data may be a little tricky, unfortunately.
- Use *git*. This is by far the hardest option to learn, but it's also by far the best option for those seeking a career in data science.

6.2 Tubric

The word “Tubric” comes from “Two-entry Rubric”. For convenience and brevity, I only describe a level 2 and a level 4 rubric item. Level 0 means it's missing/incomplete, level 1 is inadequate, and level 3 means that it was somewhere between the level 2 and level 4 items. In some cases, I include an extra level 0 condition, which means that there are some cases where you'll get an automatic 0 in that category if you meet that condition.

The grader will not read content past page 5 (not including a cover page, if you choose to include that).

There should be a table of summary statistics computed using `dplyr` and three plots made using `ggplot2`.

- **Miscellaneous**

1. Formatting

- 0: Any of the cleaned data sets are shown directly in the PDF. You can include subsets of the data to demonstrate something specific about the structure, but do not use `head()`, `glimpse()`, or simply display the data at any point in the PDF.
- 2: The PDF includes the text and the plots, possibly with some unnecessary code output or odd plot placement.
- 4: The reader makes it through the whole document without noticing the formatting because it's so natural and error-free.

2. Structure

- 2: The document follows the template, possibly with some odd changes.
- 4: The structure of the document makes navigation and reading easy, including appropriate headers and plots in reasonable places.

3. Writing Style - you can write with any level of professionalism/casualness and I am lenient with grammar, but the ideas should flow logically from one to another.

- 2: The written portions of the document have acceptable grammar and provide the correct information.
- 4: The writing is clear and concise throughout, with appropriate segues between sections. Each idea flows clearly into the next.

- **Abstract**

4. Abstract

- 2: There is an abstract that mostly puts the report in context and provides a short summary of the results.
- 4: The abstract is short and clear, and it accurately represents the entirety of the report.

- **Introduction**

5. Context - first paragraph or two should place your analysis in context.

- 2: Paragraph describes the general context.
- 4: Paragraph concisely describes the context that prepares the reader for the report without the need to reference anything in the report.

6. Introductory paragraphs - the next paragraph should provide a general idea of the data and techniques that the reader will encounter in your report.

- 2: Introductory paragraph mentions most of the techniques used.

- 4: The introductory paragraphs provide a concise overview of what the report will contain, without repeating any part of the paper verbatim.
- 7. The goal of the analysis is described.
 - 2: The goals are adequately explained. The goals can clearly be attained using the data provided.
 - 4: The goals are clearly defined. The goals can be attained using the data provided and limitations are briefly considered.
- **Data Description** - Describe each of the data sets in sufficient (but not excessive) detail.
 8. Data set 1
 - 2: The description describes the columns of the data that are relevant to the analysis.
 - 4: The description describes the information in the data concisely, possibly grouping similar columns into groups for a more concise description.
 9. Data set 2: see data set 1.
 10. Data set 3: see data set 1.
 11. Data cleaning process
 - 2: For each data set, the cleaning process is described.
 - 4: For each data set, the cleaning process is described concisely, but with enough precision that a reader could reproduce their cleaned data without seeing their code.
- **Exploration**
 12. Process description and justification
 - 2: The text describes the how and why of the data exploration
 - 4: The text is concise and fully explains and justifies the process of the exploration. No irrelevant content is included.
 13. Summary statistics table insights
 - 0: The summary statistics are only based on *one* of the data sets provided.
 - 2: The summary statistics provide insights that are relevant to the report.
 - 4: The summary statistics provide clear insights from a quick glance.
 14. Summary statistics table formatting
 - 2: The table is formatted appropriately.
 - 4: The table is exceptionally well formatted, using specialized R packages for extra fanciness.
 15. Plot 1 insights
 - 2: The plot shows something interesting.
 - 4: The insights of the plot are clear from the data display. Minimal text is required to orient the reader, and otherwise the insight stands out.
 16. Plot 2 insights: see plot 1 insights. In addition, this plot *must* have meaningful facets.
 - 0: The second plot does not have meaningful facets.
 17. Plot 3 insights: see plot 1 insights. In addition, this plot *must* draw data from two separate data sets and must be a different plot type from both of the first two (if the first two were scatterplots, this one cannot also be a scatterplot; if you have a violin and a box plot, this one must be something other than a violin or box plot).
 - 0: The third plot is based on only *one* of the data sets and is the same type as the others.
 18. Visualization quality
 - 2: The visualizations contain the relevant information, with labels and titles.
 - 4: The visualizations contain only the relevant information (with no irrelevant information), and labels and titles are used effectively.
 19. Visualization discussion - discuss the insights in the context of the data (not the general context). For example, “these data demonstrate a moderate correlation between knowledge of climate change and belief that it is happening, as measured by...” rather than “knowing about climate change means you believe that it is happening”

- 2: For each plot, the insights are discussed within the context of the
- **Conclusion**
 - 20. One-Paragraph Summary
 - 2: Paragraph summarises the report’s findings.
 - 4: The paragraph concisely summarises the reports findings and draws reasonable conclusions about the broader context.
 - 21. Future work
 - 2: Ideas for modelling/incorporating other data are discussed.
 - 4: The next steps of the model are concisely detailed, and presented in a way that excites the reader about what you might do next.
 - 22. Limitations
 - 2: The limitations of the data are discussed, with reference to the insights reported.
 - 4: The reader trusts the insights more because of the honesty in which the limitations are discussed.
- **References** (Note that links to websites are fine, but make sure that the links are relevant.)
 - 23. Insight-Related
 - 2: Less than 5 external references are used to support your insights.
 - 4: There are at least 5 references from different sources used to provide further evidence for the stated insights. The formatting makes it easy to see which reference applies to which insight. See template for example.
 - 24. Other
 - 2: There are at approximately 5 references that direct the reader to further information about the data or the techniques used.
 - 4: There are more than 5 highly relevant references that direct the reader to further information about the data and the techniques used. There are no irrelevant references, or references to help files/general documentation.
- **Bonus Marks**
 - Up to 4 marks for an especially interesting use of the provided data.
 - Up to 2 marks for using data beyond what was provided in an interesting way.
 - Up to 4 marks for proving that you used git for collaboration, with all group members contributing via commits (must be a group size of 3 or more to receive these marks; email me a link to the repo when you submit).